

Scalability of Superconductor Electronics: Limitations Imposed by AC Clock and Flux Bias Transformers

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Abstract—Flux transformers are the necessary component of all superconductor digital integrated circuits utilizing flux biasing and ac power excitation and clocking of logic cells, e.g., adiabatic quantum flux parametron (AQFP), reciprocal quantum logic (RQL), superconducting sensor arrays, qubits, etc. On average, one transformer is required per one Josephson junction. We consider limitations to the integration scale (device number density) imposed by the critical current of the ac power transmission lines (primary of the transformers) and cross-coupling between the adjacent transformers. The former sets the minimum linewidth and the mutual coupling length in the transformer, whereas the latter sets the minimum spacing between the transformers. Decreasing linewidth of superconducting (Nb) wires increases kinetic inductance of the transformer's secondary, decreasing its length and mutual coupling to the primary. This limits the minimum size. As a result, there is a minimum linewidth $w_{\min} \sim 100$ nm, which determines the maximum achievable scale of integration. Using AQFP circuits as an example, we calculate dependences of the AQFP number density on linewidth for various types of microstrip-based and stripline-based transformers and inductors available in the SFQ5ee fabrication process developed at MIT Lincoln Laboratory, and estimate the maximum circuit density as a few million AQFPs per cm^2 . We propose an advanced fabrication process for a $10\times$ increase in the density of AQFP and other ac-powered circuits. In this process, inductors are formed from a patterned bilayer of a geometrical inductance material, Nb, deposited over a layer of high kinetic inductance material, e.g., NbN. Individual patterning of the bilayer layers allows to create stripline inductors in a wide range of inductances, from the low values typical to Nb striplines to the high values typical for NbN thin films, and preserve sufficient mutual coupling in stripline transformers with extremely low cross-talk. Energy efficiency of ac-powered circuits is limited by dielectric losses in the ac power transmissions lines. Problems of scaling associated with multiphase ac power distribution are discussed.

Index Terms—Adiabatic quantum flux parametron (AQFP), cross-talk, inductance, kinetic inductance, microstrip, mutual inductance, NbN, reciprocal quantum logic (RQL), RSFQ, SFQ circuits, stripline, superconductor electronics, superconducting flux transformer, superconductor integrated circuit.

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I. INTRODUCTION

SUPERCONDUCTOR digital electronics easily beats CMOS and prospective beyond CMOS technologies in such important performance metrics as energy dissipation and processing speed. Superconductor single flux quantum (SFQ) electronics [1] hold the record in clock speed of simple circuits, about 770 GHz [2], and in its much slower, adiabatic implementations can operate with energy per bit near the Landauer's thermodynamic limit $k_B T \ln 2$ [3], [4], [5], [6]. However, these performance advantages so far have not benefited any large-scale computational system because integration scale of superconductor digital circuits is currently three to four orders of magnitude lower than that of the CMOS circuits. For instance, the largest demonstrated circuits in superconductor electronics have about one million Josephson junctions (JJs) [7], [8], whereas the modern CMOS circuits have over 50 billion transistors, a $50\,000\times$ difference [9].

Due to the recent progress in fabrication technology of niobium-based superconductor integrated circuits, the minimum feature size was reduced to 120 nm [10], [11]. This allowed for an increase in the circuit density to about $1.5\cdot 10^7$ Nb/Al-AIO_x/Nb JJs per square centimeter [10], [12], about 10-fold increase from the previous level [7]. For a comparison, the present density of CMOS circuits is $1000\times$ higher, about $1.4\cdot 10^{10}$ transistors per cm^2 [9]. The largest demonstrated density of superconductor JJ-based random access memory is 1 Mbit cm^{-2} [13]. For a comparison, RAM technology based on spin-transfer torque magnetic RAM, the so-called STT-MRAM, operating at room temperature has a $1000\times$ higher density [14], although it uses devices similar to multilayered sandwich-type JJs.

In [15], the author argued that superconductor digital electronics is fundamentally less scalable than semiconductor electronics because of the fundamental difference in information encoding. Indeed, in superconductor electronics, information is encoded, stored, and transferred by magnetic flux quanta created by superconducting currents circulating in closed superconducting loops (inductors) interrupted by JJs. In semiconductor electronics, information is encoded by a stationary electric charge on the gates of field-effect transistors. Obviously, localized charge on a capacitor occupies less space than the moving charge—superconducting loop current. Hence, charge-based devices can always be made smaller and their circuits made denser and more scaled-up than flux-based devices and circuits.

The goal of this article is to establish fundamental limits on the scalability, i.e., the maximum circuit density, of ac-powered superconductor digital electronics, imposed by two basic components of all superconductor integrated circuits: inductors and transformers. Limitations imposed by resistively shunted sandwich-type (trilayer) JJs like Nb/Al-AlO_x/Nb were discussed in [15].

A problem of superconducting transformers has emerged with advancement of ac-powered and ac-clocked superconductor logic solutions instead of the dc-powered RSFQ logic [1]. Starting from the original parametric quantron (PQ) [3], [16], [17] and going to its analogs and reincarnations—dc flux parametron [18], [19], quantum flux parametron (QFP) [20], [21], and adiabatic quantum flux parametron (AQFP) [22], [23]—all logic solutions based on parametric devices require a multiphase ac excitation. These ac signals are inductively coupled (via transformers) to the devices in order to modulate their Josephson inductance (critical current of JJs) and produce a change in the logic state and a parametrically amplified output current upon applying a weak input current I_{in} . The only exception is a dc-powered nSQUID logic whose devices, nSQUIDs, use transformers to create a large negative mutual inductance between the SQUID arms [4], [24].

Another example is reciprocal quantum logic (RQL) [25], [26], which requires four-phase ac power delivered via transformers to propagate positive and negative SFQ pulses in the same direction along Josephson transmission lines (JTLs), and provide energy to and synchronization of RQL gates.

In addition to the power and clock delivery in all types of superconductor logic and memory circuits, superconducting qubit circuits, superconducting sensor arrays, etc., superconducting flux transformers (mutual inductors) are used to provide flux biasing, signal inverting (NOT function), and coupling between cells.

As an example, for numerical simulations with well-defined parameters, we will consider what is now known as AQFP cell shown in Fig. 1(a). It is identical to QFP and PQ cells, and may only slightly differ in parameter values. The AQFP logic requires, at least, one ac transformer per JJ. Therefore, the circuit density (device number density) cannot be higher than the density of the transformers. A very similar analysis to the offered below can be easily done for RQL gates and their JTLs, which typically require one transformer per two junctions [25], [26], [27], and for any other ac-powered logic, memory, and quantum circuits.

II. AC EXCITATION AND DC FLUX TRANSFORMERS: PHYSICAL LIMITATIONS

A. Main Features of AC Excitation Transformers

The typical use of transformers in superconductor electronics is to provide a dc flux bias and/or ac flux excitation to logic cells with amplitude $\frac{\Phi_0}{\gamma}$; typically, $\gamma = 2$. Very often, e.g., in AQFP, both dc flux bias $\Phi_{dc} = \Phi_0/2$ and ac excitation are required. They can be applied either via the same or two separate transformers. For simplicity, we consider only the first case,

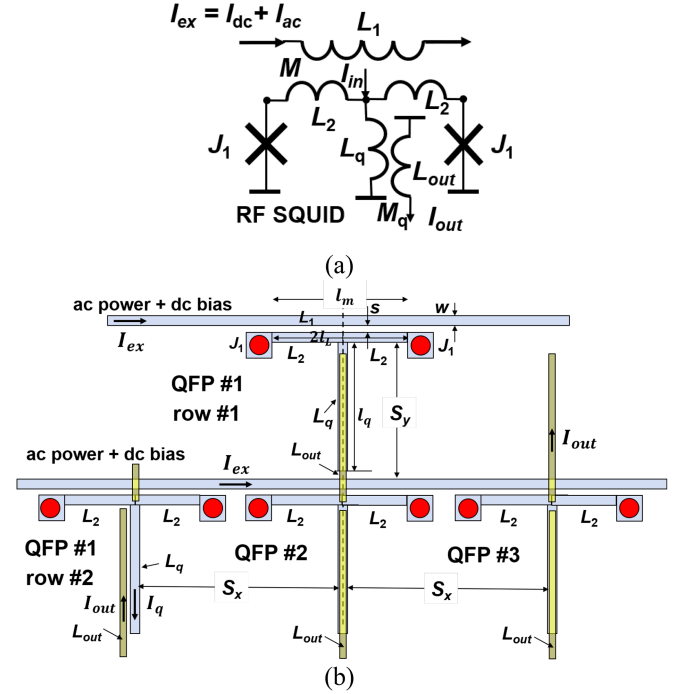


Fig. 1. (a) Schematics of PQ, QFP, and AQFP cells. It consists of two identical RF SQUIDS connected in parallel. In the adiabatic regime of operation, the typical parameters of AQFPs are $\beta_L \equiv 2\pi I_{c1} L_2 / \Phi_0 = 0.2$, $\beta_{Lq} \equiv 2\pi I_{c1} L_q / \Phi_0 = 1.6$, and $I_{c1} = 50 \mu A$ is the critical current of junctions J_1 [33], [34]. (b) Sketch of a top view of a planar transformer between the primary inductor L_1 , a part of the ac-power transmission line, and the QFP cell inductors L_2 which are either microstrip (one ground plane, not shown) or stripline (two ground planes) inductors with length l_L , laying in the same or adjacent plane as the L_1 . The mutual running length of the inductors L_1 and L_2 , which determines their total mutual inductance M , is l_m . The full QFP consists of two connected half-cells (RF SQUIDS). Also shown is the second row of QFPs. Their inputs are inductively coupled to other QFPs via the output inductor L_{out} . Spacing S_y between the QFP rows (or the ac power lines) determines their cross-coupling—ac excitation in QFP#2 produced by the power line of the QFP#1 in the same column, and vice versa. Spacing S_x determines cross-coupling between the output transformers L_q , L_{out} of QFPs in the same row. For the actual layout examples, see [37]. Inductor L_q needs to be placed perpendicular to inductors L_1 and L_2 in order to minimize direct coupling of the ac excitation to the output.

requiring $\gamma = 1$, because using two transformers increases the total transformer area by about $2\times$.

In order to provide the required dc bias and ac excitation, the mutual running length of the transformer primary wire, L_1 , and the transformer secondary wire, L_2 , should be

$$l_m = \Phi_0 / (\gamma M I_{ex}) \quad (1)$$

where $I_{ex} = I_{dc} + I_{ac}$ is the amplitude of the total excitation current that is fed into the L_1 , and M_l is the mutual inductance per unit length between the primary and secondary wires.

On the other hand, the transformer secondary, L_2 is always a part of the logic cell and its inductance is determined by the cell design parameter

$$\beta_L = 2\pi I_{c1} L_2 / \Phi_0 \quad (2)$$

where I_{cJ} is the critical current of the cell JJ. Hence, length of the inductor L_2 is given by

$$l_L = \frac{\beta_L \Phi_0}{2\pi L_{Ll} I_{cJ}} \quad (3)$$

where L_{Ll} is the inductance per unit length of the secondary.

Obviously, the transformer with the smallest area in this simple two-wire configuration can only be formed if $l_m \leq l_L$ or if

$$I_{ex} \geq \frac{2\pi L_{Ll} I_{cJ}}{\gamma \beta_L M_l}. \quad (4)$$

Note that for AQFPs shown in Fig. 1, the total length of the ac excitation transformer secondary is $2l_L$ because it consists of two connected SQUIDS. Hence, the AQFP transformer can only be formed if $l_m \leq 2l_L$. Multiturn transformers with $l_m > 2l_L$ are possible but will not be considered here because they always occupy much larger area and, hence, restrict the circuit density more than the simplest parallel-wire transformer shown in Fig. 1(b).

The mutual inductance, M , of any two conductors is always smaller than the magnetic part of the self-inductance L_M of the conductors

$$M = \kappa(L_{M1} L_{M2})^{1/2} \quad (5)$$

$$L_l = L_{Ml} + L_{Kl} \quad (6)$$

$$L_{Kl} = \mu_0 \frac{\lambda^2}{tw} \quad (7)$$

where L_{Kl} is kinetic inductance per unit length of a wire with thickness t and width w , λ is the magnetic field penetration depth, $\kappa \leq 1$ is the coupling coefficient between the magnetic inductances of the primary, and L_{M1} and the secondary, L_{M2} . If the primary and secondary are formed by inductors of the same type with equal magnetic inductance per unit length, $L_{M1l} = L_{M2l} = L_{Ml}$, the required primary excitation current from (4) is

$$I_{ex} \geq \frac{2\pi}{\gamma \beta_L \kappa} \left(1 + \frac{\mu_0 \lambda^2}{tw L_{Ml}}\right) I_{cJ}. \quad (8)$$

This current is significantly larger than I_{cJ} and *must increase with decreasing the cross-section of the inductors*, i.e., with increasing the scale of integration.

On the other hand, I_{ex} must be smaller than some maximum current $I_{\max}$ related to the critical current, I_c , of superconducting (Nb) wires forming the transformer

$$I_{ex} \leq I_{\max} = \alpha I_c = \alpha j_c w t \quad (9)$$

where $\alpha < 1$ is a safety factor and j_c is the superconductor critical current density. Hence, the minimum possible inductor cross-sectional area $A = wt$ can be found from the condition

$$wt \geq \frac{2\pi}{\alpha \beta_L \gamma j_c \kappa} \left(1 + \frac{\mu_0 \lambda^2}{tw L_{Ml}}\right) I_{cJ} \quad (10)$$

which is not a quadratic equation because of a nontrivial dependence of L_{Ml} and κ on t and w .

In superconductors, the fundamental limit to the j_c is the Ginzburg–Landau critical current density j_c^{GL} given by

$$j_c^{\text{GL}} = \frac{2\sqrt{2}}{3\sqrt{3}} \frac{B_c}{\mu_0 \lambda} \quad (11)$$

where B_c is the thermodynamic critical magnetic field of the superconductor and λ is the magnetic field penetration depth [28]. Using the microscopic theory [29], Bardeen [30] expressed the depairing critical current density as

$$j_c^{\text{GL}} \approx B_{c0} \left(\frac{\Delta_0}{\mu_0 \hbar \rho} \right)^{\frac{1}{2}} \left(1 - \frac{T^2}{T_c^2} \right)^{\frac{3}{2}} \quad (12)$$

where B_{c0} and Δ_0 are the thermodynamic critical field and the energy gap at zero temperature $T = 0$, respectively; ρ is the film resistivity in the normal state, and $\lambda(0) = \left(\frac{\hbar \rho}{\mu_0 \pi \Delta_0} \right)^{1/2}$.

For niobium, $B_c(T) = B_{c0}(1 - T^2/T_c^2)$, $T_c = 9.1$ K, and $B_{c0} = 0.155$ T, giving $B_c = 0.122$ T at 4.2 K. The measured value of the penetration depth in our Nb films at 4.2 K is $\lambda = 90$ nm. Then, (11) gives $j_c^{\text{GL}}(4.2 \text{ K}) = 0.59 \text{ A}/\mu\text{m}^2$ for our Nb films at 4.2 K, which agrees with the value following from (12) and the measured values of Δ_0 and residual ρ in our films. The measured critical current density j_c in Nb wires with $w \approx t \approx 200$ nm is somewhat lower than this theoretical value, about $0.37 \text{ A}/\mu\text{m}^2$ [10], [31]. Since j_c may decrease further with reducing the wire dimensions, e.g., due to increasing Nb contamination and resistivity, it would be practical to take $\alpha j_c = 0.25 \text{ A}/\mu\text{m}^2$, a 33% lower value than the measured critical current density, in order to have some safety margin.

To proceed further with (10) and find the dependence of the transformer area on the linewidth, we need to specify the type of the transformer in order to determine L_{Ml} and κ . Before going to numerical results, as a simple illustration, we consider a transformer formed by two parallel microstrips laying in the same plane.

For superconducting microstrips with rectangular cross-section, the magnetic part of the inductance per unit length is given in [32] by

$$L_{Ml} = \frac{\mu \mu_0}{4\pi} \ln \left[1 + \frac{4(d_1 + \frac{t}{2} + \lambda)^2}{0.2235^2 (w + t)^2} \right] \quad (13)$$

where d_1 is the dielectric thickness between the microstrip and the ground plane, and μ is the magnetic permeability of the dielectric, hereafter assumed to be 1. From the fabrication process practicality, we are mostly interested in narrow wires with deeply scaled features $w \approx t$ and $w, t \ll 2(d_1 + \lambda)$. In this case, the minimum cross-section area is given by solution of the equation

$$A \geq \frac{2\pi}{\alpha \beta_L \gamma j_c \kappa} \left(1 + \frac{4\pi \lambda^2}{A \mu \ln(1 + \frac{(d_1 + \lambda + t/2)^2}{0.05 A})} \right) I_{cJ}. \quad (14)$$

To solve (14), let us use the typical parameters of ac excitation and flux bias transformers in AQFPs as an example: $I_{cJ} = 50 \text{ }\mu\text{A}$, $\gamma = 1$, and $\beta_L = 0.2$ [22], [33], [34]. In the widely used fabrication process SFQ5ee [35], transformers using Nb microstrips M6aM4 (standing for microstrips with the signal

trace on the layer M6 above the M4 ground plane in [35]) with $d_1 = 615$ nm are the most convenient because they are the closest to the layer of JJs. Solution of (14) at $\kappa = 1$ gives the smallest cross-section area of the primary wire $A_{\min} = 1.34 \cdot 10^{-2} \mu\text{m}^2$, corresponding to the minimum linewidth and the film thickness of about $(A_{\min})^{\frac{1}{2}} \approx 116$ nm. *At smaller cross-sections, the required transformer cannot be made because the excitation current required to induce Φ_0 flux amplitude in the transformer secondary would exceed the critical current of the transformer primary wire.* Achieving the largest coupling $\kappa = 1$ in the transformer is not possible. For a more realistic $\kappa = 0.5$, solution of (14) gives $A_{\min} = 2.18 \cdot 10^{-2} \mu\text{m}^2$ and $(A_{\min})^{\frac{1}{2}} = 148$ nm.

The minimum linewidth following from the minimum cross-section area determines the minimum possible coupling length between the transformer primary and secondary, i.e., the minimum possible size of the ac-powered cell along the ac power transmission line. Existence of the minimum linewidth and of the minimum cell size is the first limit on scalability of ac-powered superconductor electronics (AQFP, RQL, etc.) caused by the finite superconducting critical current of the primary wire in the ac and dc flux-biasing transformers. This limit can be reached already in a 90-nm technology node and 90-nm linewidth. Further reduction of the linewidth would not significantly increase the density of superconductor integrated circuits using ac powering of logic gates. This is our first conclusion.

B. Output Coupling Transformer in AQFP and Inductor L_q

In this subsection, we consider limitations on the cell height (in the y-direction perpendicular to the ac power transmission line) and width. For specificity, we use the typical parameters of AQFPs, whereas the same arguments and estimates apply to other types of ac-powered superconductor logics.

Length of the inductor L_q in AQFP in Fig. 1, l_q , is given by

$$l_q = \frac{L_q}{L_{ql}} = \frac{\beta_{Lq} \Phi_0}{2\pi L_{ql} I_{cJ}} \quad (15)$$

where L_{ql} is inductance per unit length of inductor L_q , which may differ from the per length inductance of the excitation transformer secondary L_{Ll} . At the typical AQFP parameters, $I_{cJ} = 50 \mu\text{A}$ and $\beta_{Lq} = 1.6$, L_q is 10.53 pH; at $L_{ql} \sim 1 \text{ pH}/\mu\text{m}$, l_q is about 10 μm .

Inductor L_q needs to be placed perpendicular to the inductors L_1 and L_2 , forming a T-shape, in order to minimize direct coupling of the ac excitation to the output. The aspect ratio of this "T" in the typical AQFP cell (width to length ratio) is $2l_L/l_q = 1:4$ because the ratio of the optimal parameters $\beta_q/\beta_L = l_q/l_L$ is 1.6:0.2 = 8:1.

Consider inductive coupling with mutual inductance M_q between the L_q and the output inductor L_{out} connected to the next AQFP or forming a part of the buffer which sums up the output currents of three AQFPs comprising the majority (MAJ3) gate [37]. Parametrically amplified input current I_{in} creates a current

$$I_q \approx \Phi_0/L_q = 2\pi I_{cJ}/\beta_q \approx 200 \mu\text{A} \quad (16a)$$

in the L_q , which in turn creates a current

$$I_{\text{out}} = M_q I_q / L_{\text{out}} \approx 2\pi M_q I_{cJ} / (\beta_q L_{\text{out}}) \quad (16b)$$

in the output inductor. The maximum value of L_{out} can be determined from the condition $I_{\text{out}} \geq I_{\text{in}} \approx 10 \mu\text{A}$, which is the current required to drive the next AQFP. This gives

$$L_{\text{out}} \leq \frac{2\pi M_q I_{cJ}}{\beta_q I_{\text{in}}} = 19.63 M_q \quad (17a)$$

$$l_{\text{out}} = \frac{19.63 M_q}{L_{\text{outl}}} \quad (17b)$$

The latter determines the maximum distance over which the AQFP output data can be sent over without amplification, i.e., the maximum length of the inductor L_{out} ; L_{outl} is inductance per unit length of the output inductor. Using $M_{ql} \sim 0.3 \text{ pH}/\mu\text{m}$, $l_q = 10 \mu\text{m}$ and $L_{\text{outl}} \sim 1 \text{ pH}/\mu\text{m}$, we get $l_{\text{out}} \sim 60 \mu\text{m}$, about $6l_q$. Decreasing L_{outl} by increasing the output inductor width may help in transmitting data to $10\times$ larger distances than the height of the AQFP cell.

The superconducting material and cross-section of wires for inductors L_q and L_{out} are selected such that both currents I_q (16a) and I_{out} (16b) are smaller than the critical current of the corresponding wires. For Nb wires with $\alpha j_c = 0.25 \text{ A}/\mu\text{m}^2$, the minimum cross-section of L_q is $A_{\min} = 8 \cdot 10^{-4} \mu\text{m}^2$, which is an order of magnitude smaller than $A_{\min}$ following from (14) and corresponds to the wire dimensions $(A_{\min})^{\frac{1}{2}} \sim 30$ nm. This is the ultimate limit to reduction of the linewidth of Nb wires.

C. QFP Minimum Area and Maximum Circuit Density

The minimum possible area of the typical QFP is $A_{\text{QFP}} = 2l_L l_q = \frac{\beta_L \beta_q \Phi_0^2}{2\pi^2 L_{Ll} L_{ql} I_{cJ}^2}$ if the area occupied by the junctions J_1 , the transformer primary, and vias between the inductors can be neglected. If the JJs cannot be hidden inside this rectangle, their area, which in the simplest case is $A_{JJ} = 2I_{cJ}/J_c$, needs to be added; J_c is the Josephson critical current density of the junctions, e.g., $J_c = 100 \mu\text{A}/\mu\text{m}^2$ in the SFQ5ee fabrication process [35]. Then, the QFP maximum number density is

$$n_{\text{QFP}} = \chi / (A_{\text{AQFP}} + A_{JJ}) = \frac{\chi}{\frac{\beta_L \beta_q \Phi_0^2}{2\pi^2 L_{Ll} L_{ql} I_{cJ}^2} + \frac{2I_{cJ}}{J_c}}, \quad (18a)$$

where $\chi < 1$ is the area filling factor. The density (18a) is maximized at

$$I_{cJ}^{\text{opt}} = \left(\frac{\beta_L \beta_q J_c \Phi_0^2}{2\pi^2 L_{Ll} L_{ql}} \right)^{1/3}. \quad (18b)$$

Using parameters of the optimized AQFP cells, $\beta_L = 0.2$ and $\beta_q = 1.6$ [33], [34], [39], $J_c = 100 \mu\text{A}/\mu\text{m}^2$, and the maximum expected value $L_{Ll} = L_{ql} \approx 1 \text{ pH}/\mu\text{m}$ for Nb inductors, we get $I_{cJ}^{\text{opt}} = 191 \mu\text{A}$ and the maximum density of parametrons $n_{\text{QFP}}(I_{cJ}^{\text{opt}}) = 1.75 \cdot 10^7 \text{ cm}^{-2}$ at $\chi = 1$. Unfortunately, this I_{cJ}^{opt} is too high for using in AQFPs from the standpoint of energy dissipation, which is proportional to I_{cJ} . To provide ultralow dissipation in these adiabatic devices, I_{cJ} is reduced as much as possible to a level set by an acceptable bit error rate (BER) in complex circuits; BER grows exponentially with I_{cJ} reduction. The most typical value is $I_{cJ} = 50 \mu\text{A}$ [33], [34], [39], a factor of four lower than I_{cJ}^{opt} .

At $I_{cJ} = 50 \mu\text{A}$, (18a) gives $n_{\text{QFP}} = 3.5 \cdot 10^6 \text{ cm}^{-2}$ at $\chi = 1$. This is our second conclusion—the number density of AQFPs using Nb inductors is limited from above to about three million AQFPs per cm^2 , corresponding to about $1 \cdot 10^6 \text{ cm}^{-2}$ density of AQFP majority gates composed of three (MAJ3) AQFPs. Increasing the Josephson critical current density to $J_c = 600 \mu\text{A}/\mu\text{m}^2$, which is available in MIT LL fabrication processes [36], [37], has negligible effect on n_{QFP} since the device density at low I_{cJ} values is fully determined by the AQFP inductors; see (18a).

The n_{QFP} estimate above does not account for possible limitations caused by a finite supercurrent-carrying capacity of the ac power transformer and for a possibility of using inductors with higher L_{ql} values. Accounting for these factors requires a detailed analysis given in Section III but does not change the order of magnitude of the maximum density of AQFPs.

Increasing L_{ql} values significantly above $1 \text{ pH}/\mu\text{m}$ can be done using kinetic inductance (7). For instance, a 40-nm thick Mo_2N film used in the SFQ5ee process [35] has $L_{Kl} \approx 8/w \text{ pH}/\mu\text{m}$ (w is in micrometers) [38], which is $\gg 1 \text{ pH}/\mu\text{m}$ at $w < 1 \mu\text{m}$. However, a short strip of kinetic inductor can be used only if L_q is galvanically coupled to the next AQFP, the so-called directly coupled AQFP [40]. If inductive coupling to the next AQFP is required, the mutual inductance M_q between the short strip of a kinetic inductor and the output inductor L_{out} , which is proportional to the length of L_q , is going to be small. Careful optimization is needed in this case, as will be discussed in Section III. Also, j_c of kinetic inductors is much smaller than j_c of Nb because of a much larger λ in (13). For instance, if the L_q is made of a 40-nm-thick Mo_2N kinetic inductor in the SFQ5ee process, its critical current would be reached at $w \leq 0.5 \mu\text{m}$ [38]. Nevertheless, implementation of specially optimized kinetic inductors in AQFPs, and in superconductor electronics in general, could substantially decrease the size of logic cells and increase their number density [15].

D. Cross-Coupling of Transformers and Circuit Density

Another factor limiting the practical density of transformers and ac-powered cells is cross-talk between them. AC-powered cells are arranged along transmission lines forming a power grid feeding ac power into them, as shown schematically in Fig. 1(b) for QFPs; see also [34] and [39]. Cells arranged along one horizontal power transmission line induce some ac excitation in the cells coupled to the adjacent transmission lines of the grid. The minimum distance $S_{\min}$ between the adjacent rows of the ac-powered cells is determined by the acceptable level of cross-coupling between their transformers. We define this cross-coupling as the ratio of the mutual inductance M_{cross} between the primary L_1 of the transformer in one row and the secondary L_2 of the transformer in the adjacent row to the mutual M inductance within the same transformer. This ratio determines the ac excitation amplitude induced by a transmission line feeding one row of the cells, e.g., QFPs, into the adjacent rows of the cells below and above.

In a given fabrication process, the minimum practical spacing between the adjacent rows of QFPs is set by the $\max[l_q, S_{\min}]$,

where $S_{\min}$ is the spacing at which the maximum acceptable level of cross-talk is reached. As an example, we will hereafter use 5% cross-talk as this maximum acceptable level. Reducing the length l_q below the $S_{\min}$ would not increase n_{QFP} . Hence, $S_{\min}$ sets the maximum value of L_{ql} required to maximize the density of QFPs, e.g., by using kinetic inductors

$$L_{ql} = \frac{\beta_{Lq} \Phi_0}{2\pi S_{\min} I_{cJ}} \approx \frac{10.5}{S_{\min}} \text{ (in pH}/\mu\text{m}) \quad (19)$$

where $S_{\min}$ is in micrometers.

We also need to consider cross-coupling between the adjacent QFPs in the same row, i.e., coupling between parallel inductors L_q and L_{out} in two adjacent QFPs. Current I_{q1} in QFP #1 induces flux $\Phi_{21} = M_{q21} I_{q1}$ in the adjacent QFP #2 and current $I_{\text{out}21}$ in its output inductor $L_{\text{out}2}$. Reliable operation of the circuit may require this flux to be small in comparison with flux Φ_0 created by ac excitation in QFP #2 and the induced current to be small in comparison to self-current $I_{\text{out}22}$ (16) at the output of QFP #2. As in the case of ac excitation transformers, we define cross-coupling between the output transformers as

$$i_{21} = \frac{I_{\text{out}21}}{I_{\text{out}22}} = \frac{M_{q21}}{M_q}. \quad (20)$$

Its maximum tolerable value sets the minimum spacing S_q between inductors L_q and L_{out} in the adjacent QFPs in the same row along the ac power transmission line.

III. MICROSTRIP TRANSFORMERS AND QFP NUMBER DENSITY

For more accurate estimates of the number density of the ac transformers and QFPs, in the following sections, we will consider all types of possible transformers with the goal to minimize the area of the QFP cell.

A. QFP Transformers Formed by Two Parallel Microstrips in One Plane

Mutual inductance of two microstrips in the same plane decreases slowly with distance between the centers of their cross-sections, $p_x = w + s$, and is given in [32] by

$$M_l = \frac{\mu\mu_0}{4\pi} \ln \left[1 + \frac{4(d_1 + \lambda + \frac{t}{2})^2}{(w + s)^2} \right]. \quad (21)$$

For definiteness, we take parameters of the most advanced fabrication processes for superconductor electronics: the SFQ5ee process [35]. These parameters and the process cross-section are given in [Table I, 11] and [Fig. 1, 32]. Nb wires are on the process layer M6 with thickness $t = 200 \text{ nm}$, 200-nm thick Nb ground plane is layer M4, and the dielectric thickness between them is $d_1 = 615 \text{ nm}$. The currently allowed minimum linewidth and spacing between the microstrips in the same plane is 250 nm . Hopefully, with the progress of fabrication technology, narrower lines and gaps between metal lines with dielectric fill will become possible, e.g., due to the development of a damascene-type processing allowing for much smaller spacings $s \sim w$.

Fig. 2 shows the mutual inductance per unit length of two microstrips M6 above M4 ground plane (M6aM4) as a function of their linewidth at spacing $s = 250 \text{ nm}$ and the microstrip

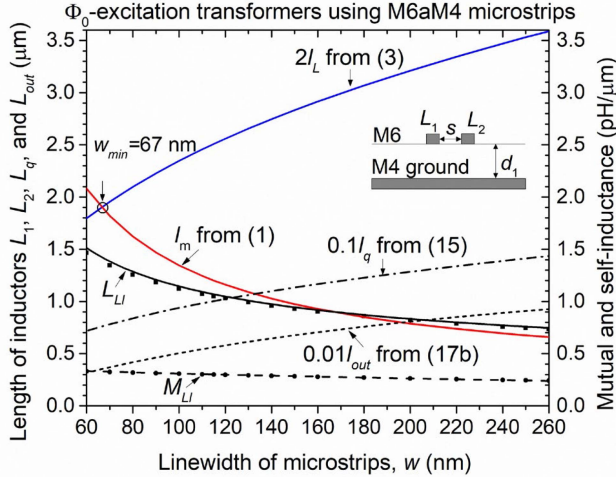


Fig. 2. Parameters of Φ_0 -excitation in-plane transformers using microstrips M6 above M4 ground plane (M6aM4), spaced at $s = 250$ nm in the SFQ5ee process; the transformers cross-section is sketched in the inset. Mutual inductance of the two microstrips per unit length, M_L (right scale): black dash (21); (•)—numerical simulations using wxLC [41]. Self-inductance per unit length, L_L (right scale): black solid curve is the sum of magnetic (13) and kinetic (17) inductances; (■)—numerical simulations using wxLC. Thicknesses of Nb layers forming the microstrips: $t_{M6} = 200$ nm, M4 ground plane $t_{M4} = 200$ nm; dielectric thickness $d_1 = 615$ nm, $\lambda = 90$ nm. Solid red curve is the mutual running length l_m (1) required to induce Φ_0 in the AQFP secondary formed by two inductors L_2 , assuming the maximum allowed current density of 0.25 A/ μm^2 in the primary inductors L_1 . The top solid blue curve is the length $2l_L$ (3) of the two inductors L_2 in the AQFP cell at $I_{c,J} = 50$ μA and $\beta_L = 0.2$. Below the $w_{\min} \approx 70$ nm, corresponding to the circled intersection, $l_m > 2l_L$ and transformers with the required mutual inductance M cannot be formed. Dash-dot curve is $0.1l_q$; l_q is the length of M6aM4 microstrip inductor L_q with the same width as the L_1 and L_2 ; at $\beta_{Lq} = 1.6$, $l_q = 8l_L$. Short dash is $0.01l_{\text{out}}$; l_{out} is the maximum length (17b) of the output inductor L_{out} , assuming it is also a M6aM4 microstrip with the same width w , coupled at $s = 250$ nm to the microstrip L_q along the full length of the L_q .

self-inductance per unit length (dash curve, scale on the right). It also shows the required mutual running length l_m in (1) of the primary L_1 , the length $2l_L$ in (3) of the secondary formed by two inductors L_2 (solid red curve). To show in the same scale, we also plotted $1/10^{\text{th}}$ of the l_q given by (15) and $1/100^{\text{th}}$ of the l_{out} given by (17b), assuming that the output transformer is also formed by two M6aM4 microstrips spaced at $s = 250$ nm, i.e., that the output transformer is of the of same design as the ac excitation transformer. In addition to analytical expressions given in [32], we used a very accurate numerical inductance extractor wxLC developed by M. Khapaev [41] and $\lambda = 90$ nm for all Nb layers. The analytical and numerical, shown by solid symbols, results are practically indistinguishable.

Below the linewidth $w_{\min} \approx 67$ nm, the required mutual running length of the ac excitation line l_m needs to be longer than the total length $2l_L$ of the QFP inductors L_2 in order to provide a Φ_0 sum of the dc flux bias and ac excitation amplitude without exceeding the maximum allowed current $I_{\max} = \alpha j_c w_{\min} t_{M6} = 3.5$ mA in the primary. *The parallel-line microstrip transformer with the required parameters cannot be formed using smaller linewidths.* The minimum cross-section area of the primary $w_{\min} t_{M6} = 0.0134$ μm^2 perfectly agrees with the $A_{\min}$ estimated using (14) in Section II-A.

For hypothetical transformers with $s = w$, the excitation current limit is reached below $w_{\min} \approx 50$ nm, corresponding to $I_{\max} \approx 2.5$ mA, due to a stronger mutual coupling in the transformer.

In practical circuits, we want to reduce the ac excitation current I_{ex} in the primary as much as possible in order to minimize ac power loss caused by dielectric losses in the transmission lines, which grows as I_{ex}^2 . The minimum I_{ex} is obtained at the largest possible mutual running length at a given linewidth, i.e., at $l_m = 2l_L$. In this case, the full QFP transformer consisting of two parallel wires in the same plane has area $A_{\text{tr}} = 2l_L(2w + s)$, where l_L is given by (3). The minimum size of the rectangular QFP cell for tiling is $(2l_L + s) \times (2w + 2s + l_q)$, where $(2l_L + s)$ is the minimum possible tiling pitch in the x -direction along the ac power transmission line, and $(2w + 2s + l_q)$ is the tiling pitch (effective height of the QFP cell) in the y -direction.

B. Cross-Coupling of Planar Microstrip-Based Transformers and the Limits on QFP Number Density

To mitigate cross-coupling between two parallel transformers in the adjacent rows of QFPs, e.g., shown in Fig. 1(b), the spacing S_y between the primary of transformer #1 and the secondary of transformer #2 should be much larger than the spacing s between the wires in the transformer. The cross-coupling $M(S, w)/M(s, w)$ between two M6aM4 microstrip-based planar transformers is shown in Fig. 3(a) for a few linewidths, where $M(s, w)$ is given by (21) at $s = 250$ nm and $M(S, w)$ is given by (21) with S replacing s . Numerical simulations using wxLC [41] are shown by (•) and perfectly agree with (21).

We see that providing a low cross-talk between the in-plane microstrip transformers requires quite large spacings, e.g., cross-talk below 10% requires $S > 2.5$ μm and below 5% requires $S \geq 3.75$ μm . The cross-talk increases with increasing w and decreases with decreasing s .

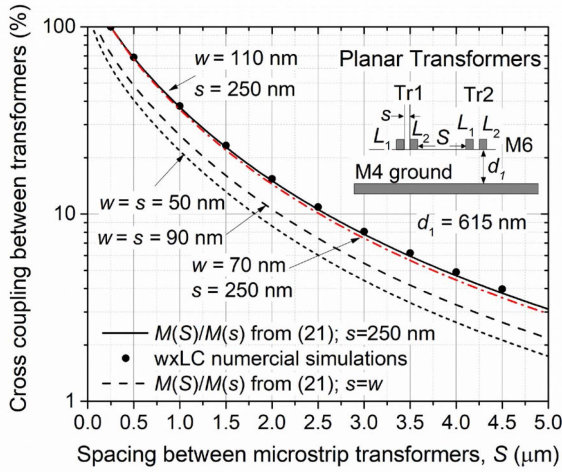
The maximum acceptable cross-talk level depends on the circuit. For specificity, we hereafter take 5% as the maximum acceptable cross-talk level. This defines the minimum spacing between the transformers, $S_{\min}(w, s)$, a function of the in-transformer linewidth and spacing. This function is shown in Fig. 3(b) for the planar M6aM4 microstrip transformers with $s = 250$ nm (solid blue curve) in the SFQ5ee process and for the transformers with $w = s$ (red dash curve).

If we restrict the spacing S_y between the horizontal rows of the QFPs to the largest of the $l_q + s$ and $S_{\min}$ in order to keep cross-talk between the ac excitation transformers at the acceptable level, the number density of the QFPs becomes

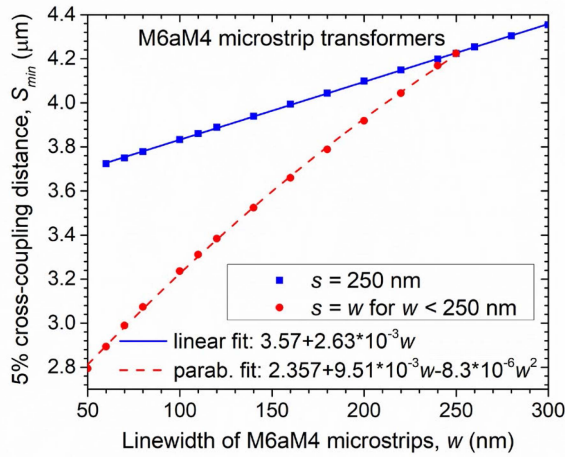
$$n_{\text{QFP}} = \frac{\chi}{(2l_L + s)(2w + 2s + l_q)}, \text{ if } l_q + s \geq S_{\min}, w \geq w_{\min} \quad (22a)$$

where $\chi \leq 1$ is the area filling factor.

Linewidths $w < w_{\min}$ cannot be used in the QFP ac excitation transformers because of the critical current limitation, *but can be used for the inductor L_q* . In this case, the cell height continues to decrease because reduction of the inductor L_q width, w_q , reduces l_q , whereas the cell effective width $(2l_L + s)$ remains constant and equal $2l_L$ (at $w_{\min}$) + s . Hence, if $w < w_{\min}$ and



(a)



(b)

Fig. 3. (a) Cross-coupling $M(S, w)/M(s, w)$ between in-plane (planar) microstrip-based transformers with various linewidths and in-transformer spacings, s , as a function of spacing S between the adjacent transformers. All the curves were calculated using (21) for the transformers using M6aM4 microstrips; (•)—numerical simulations using wxLC software [41] for $w = 110$ nm. (b) Dependence of the spacing $S_{\min}$ at which mutual inductance between the transformers reduces to 5% of the in-transformer mutual inductance, referred to as a 5% cross-coupling distance, on the linewidth: blue squares—transformers with the primary to secondary spacing $s = 250$ nm; red dots—transformers in a hypothetical process with $s = w$ for $w \leq 250$ nm. Solid blue line and red dash curve are, respectively, a linear fit $S_{\min}(w, s) = 3.57 + 2.63 \times 10^{-3}w$ and parabolic fit $S_{\min}(w, s) = 2.357 + 9.51 \times 10^{-3}w - 8.3 \times 10^{-6}w^2$ in these two cases; here $S_{\min}$ is in micrometers and w is in nanometers.

$l_q + s \geq S_{\min}$, the number density of QFPs is given by

$$n_{QFP} = \frac{\chi}{(2l_L|_{w=w_{\min}} + s)(2w_{\min} + 2s + l_q)}, \text{ if } l_q + s \geq S_{\min}, w < w_{\min}. \quad (22b)$$

In the opposite case $S_{\min} > l_q + s$, the number density is given by

$$n_{QFP} = \frac{\chi}{(2l_L + s)(2w + s + S_{\min})}, \text{ if } S_{\min} \geq l_q + s,$$

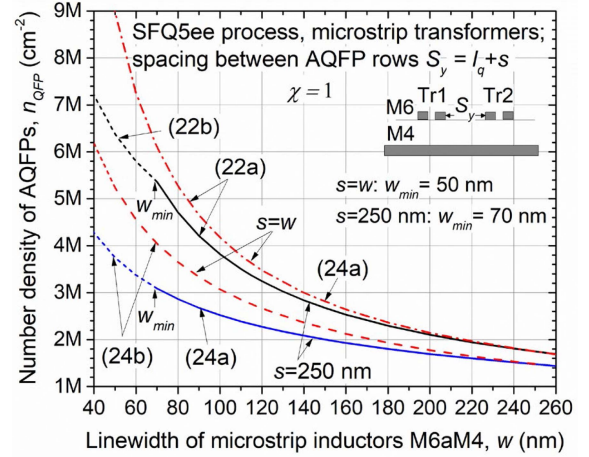


Fig. 4. Theoretical number density (assuming 100% area filling, $\chi = 1$) of AQFPs using M6aM4 microstrip inductors and transformers in the SFQ5ee process with parameters of Nb inductors L_1 , L_2 , and L_q given in Fig. 2 for the AQFP cell in Fig. 1. Inset shows the cross-section of the AQFP transformers. The bottom blue solid ($w > w_{\min}$) and short dash ($w < w_{\min}$) curves correspond to $s = 250$ nm and spacing S_x between the AQFP in the x -direction set by the 5% cross-coupling distance $S_q(w, s)$ because $2l_L + s < S_q(w, s)$ at all linewidths, whereas spacing S_y in the y -direction is set by the size of the AQFP cell, $l_q + s$. The black solid and short dash curves correspond to the n_{QFP} determined only by the physical dimensions of the AQFPs, respectively (22a) and (22b), ignoring cross-coupling. The uppermost red dash-dot curve is (24) for a hypothetical process with $s = w$ and the physical dimensions of the AQFP, ignoring cross-coupling between AQFPs in the same horizontal row. The red dash and short dash curves in the middle also correspond to $s = w$ but account for the cross-coupling in the same row (24b), using the dependence $S_q(w, s) = S_{\min}(w, s)$ in Fig. 3(b); see text. Cross-coupling between the adjacent rows in the y -direction can be neglected in all the cases because $l_q + s > S_{\min}(w, s)$ at all the linewidths.

$$w \geq w_{\min} \quad (23a)$$

$$n_{QFP} = \frac{\chi}{(2l_L|_{w=w_{\min}} + s)(2w_{\min} + s + S_{\min})}, \text{ if } S_{\min} \geq l_q + s \text{ and } w < w_{\min}. \quad (23b)$$

Using Nb microstrips M6aM4 as AQFP inductors in the SFQ5ee process gives $l_q \geq S_{\min}$ for all linewidths down to about 20 nm. Hence, (22a) applies if we ignore cross-coupling between adjacent QFPs in the same row; see below. Dependences (22a) and (22b) for QFPs with the M6aM4 inductors and transformers with $s = 250$ nm are shown in Fig. 4 by the solid black ($w \geq w_{\min}$) and black dash ($w < w_{\min}$) curves.

So far, we have ignored cross-talk between the data output transformers L_q , L_{out} of adjacent QFPs in the same horizontal row. However, placing adjacent QFPs in the same row at the minimum spacing $s = 250$ nm gives the horizontal spacing S_x between inductors L_q in the range $2.75 \leq 2l_L + s \leq 3.85$ μm . At these distances, the cross-talk between the parallel output inductors can be very substantial as follows from (21) and Fig. 3(a). So the QFPs may need to be placed further apart in the x -direction (along the ac transmission line) and spaced at some distance S_q to reduce the output cross-talk to the acceptable level. Then, (22a) and (22b) are replaced by,

respectively, following equations:

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w)(2w + 2s + l_q)}, \text{ if } l_q + s \geq S_{\min},$$

$$S_q \geq 2l_L + s, \text{ and } w \geq w_{\min} \quad (24a)$$

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w)(2w_{\min} + 2s + l_q)}, \text{ if } w < w_{\min},$$

$$l_q + s \geq S_{\min}, \text{ and } S_q \geq 2l_L + s \quad (24b)$$

where we use w also for the width of the inductors L_q .

If the minimum spacing of QFPs in both the x - and y -directions are set by the acceptable cross-coupling, then

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w)(2w + s + S_{\min})}, \text{ if } l_q + s < S_{\min}, S_q > 2l_L$$

$$+ s, \text{ and } w \geq w_{\min} \quad (25a)$$

$$n_{\text{AQFP}} = \frac{\chi}{(S_q + w)(2w_{\min} + s + S_{\min})}, \text{ if } l_q$$

$$+ s < S_{\min}, S_q > 2l_L + s, \text{ and } w < w_{\min}. \quad (25b)$$

If the acceptable cross-talk level between the adjacent inductors L_q is the same as between the excitation transformers, e.g., less than 5%, then $S_q = S_{\min}$ because the output inductors are of the same type as in the ac excitation transformers. Then, the minimum pitch of AQFP placement in the same row is $S_{\min} + w$, and the number density of AQFPs using the planar M6aM4 transformers is given by (24a) and (24b). The corresponding number density of AQFPs is shown in Fig. 4 by the lowest blue solid (24a) and short-dash (24b) curves calculated using the dependences in Fig. 3(b) and $S_q(w, s) = S_{\min}(w, s)$.

For the currently available SFQ5ee process with $s = 250$ nm and already demonstrated linewidth $w \approx 110$ nm [10], [11], the theoretical AQFP number density is about $3.5 \cdot 10^6 \text{ cm}^{-2}$, set by the physical dimensions of the AQFP and ignoring in-row cross-talk. This numerical analysis agrees perfectly with the simple estimate obtained in Section II-C and based on (18a). The maximum density reduces to about $2.5 \cdot 10^6 \text{ cm}^{-2}$ if cross-talk needs to be kept below the 5% level. These densities correspond to, respectively, 1.17M and 0.83M MAJ3 logic gates per cm^2 . We emphasize that the considered AQFPs are about 30 times smaller in area than $30 \mu\text{m} \times 40 \mu\text{m}$ AQFP buffer cells used in [39].

Reducing spacing between superconducting lines below 250 nm in a hypothetical future development of the SFQ5ee process would bring some increase to the device density, as follows from (21)–(24) and Fig. 3(b). For instance, using $s = w$ would increase mutual coupling in the planar transformer, resulting in $w_{\min} \approx 50$ nm. Cross-coupling between the microstrip transformers would also noticeably decrease, resulting in $S_q(5\%) = S_{\min}(5\%) \approx 2.75 \mu\text{m}$ at $w = 50$ nm, as shown in Fig. 3(b). This minimum spacing can be used to estimate the absolutely maximum possible value of n_{QFP} in this technology.

Even at $w = 40$ nm, $l_q = 5.66 \mu\text{m}$ is still twice as large as the $S_{\min}$, and the n_{QFP} is still given by (22a) and (22b), as shown in Fig. 4 by the uppermost dash-dot red curve, reaching about $9 \cdot 10^6 \text{ cm}^{-2}$ at $w = 50$ nm. If the placement pitch S_x of AQFPs

in the same row is set by the 5% cross-talk requirement between the output transformers $S_q = S_{\min}(w, s)$, the n_{QFP} is given by (24a) and (24b), and shown in Fig. 4 by the red dash ($w \geq w_{\min}$) and short dash ($w < w_{\min}$) curves, respectively, reaching about 6.2M cm^{-2} at $w = 40$ nm.

For a more realistic 90-nm process ($w = s = 90$ nm), $n_{\text{QFP}} = 4.65 \cdot 10^6 \text{ cm}^{-2}$ and $3.80 \cdot 10^6 \text{ cm}^{-2}$ without and with accounting for the in-row cross-talk, respectively. This is less than 33% increase over the density of AQFPs at $w = 90$ nm and $s = 250$ nm, which would hardly justify a very complex process development required for achieving the much smaller spacing $s = 90$ nm.

C. AQFPs With M6aM4 Microstrips and Kinetic Inductor L_q

The main density limiter of AQFPs in the SFQ5ee process is a very large length of the inductor L_q . It is determined by the β_q value, which is set by the optimization of the range of adiabatic switching between the two flux states encoding information in AQFPs; see, e.g., [34]. Let us assume that in a hypothetical, next generation process, we can introduce an additional layer of kinetic inductors close to the layer of JJs and use it to make $l_q + s \leq S_{\min}(w, s)$, the length set only by the cross-talk requirements. This would bring the AQFPs in the regime (23) or (25). The only apparent drawback of reducing l_q significantly is a proportional reduction of the mutual inductance M_q in the output transformer between L_q and L_{out} , and of the data transfer length l_{out} in (17b). This reduction can be partially compensated by increasing the width of the output inductor and thereby decreasing its inductance per unit length L_{out} in (17b).

Using (19), the optimum value of the linear inductance of the kinetic inductance material replacing Nb in the M6aM4 microstrips for L_q would be $L_{ql} \approx 2.8 \text{ pH}/\mu\text{m}$, a factor of 4 higher value than the linear inductance of 250-nm-wide Nb microstrips M6aM4. For the reasonable width of the L_q , $w_q = 250$ nm, the required sheet inductance is $0.7 \text{ pH}/\text{sq}$, a factor of 10 lower than the sheet inductance of the 40-nm Mo_2N films [38] currently used in the SFQ5ee process as rf choke kinetic inductors for biasing ERSFQ circuits [42].

If in the hypothetical fabrication process we preserve somehow Nb microstrips M6aM4 in the ac excitation transformer, the n_{QFP} could be increased up to the limit set only by the acceptable cross-talk between the AQFPs. For this, the pitch of the AQFP placement in y -direction should be $2w + s + S_{\min}$, and the pitch in the x -direction should be $S_x = S_q + w_q$, where w_q is the width of the kinetic inductor L_q , because in the entire range of linewidths in Fig. 4, the physical width of the AQFP transformers $2l_L + s$ is less than $S_q(5\%)$. Due to the critical current limitation, the minimum expected value of w_q is about 200 nm. In the described case, the expected number density of AQFPs using a kinetic inductor L_q depends very weakly on the w and w_q as can be seen from the lowest black curve in Fig. 5. Below $w_{\min} = 50$ nm, n_{QFP} in this regime does not increase because the height of the AQFP cell is set by the $S_{\min}(w_{\min}, s)$ and no longer depends on the linewidth. The saturation value $(n_{\text{QFP}})_{\text{max}} = 5.6 \cdot 10^6 \text{ cm}^{-2}$ is larger than the number density which could be achieved using the ultranarrow Nb wires; see the

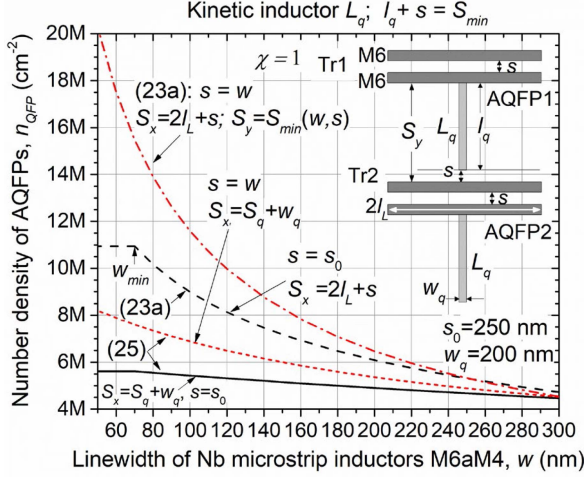


Fig. 5. Theoretical number density (assuming 100% area filling, $\chi = 1$) of AQFPs n_{QFP} in a hypothetical process using Nb microstrip inductors M6aM4 in the ac excitation transformers and a kinetic inductor L_q with width $w_q = 200$ nm and length $l_q + s = S_{min}(w, s)$, where $S_{min}(w, s)$ is a 5% cross-coupling spacing between the AQFP primary in one row and the AQFP secondary in the adjacent row shown in Fig. 3(b). Inset: a layout sketch, aerial view, of the inductors in two AQFPs located in the adjacent rows running along transmission lines in the horizontal x -direction. The bottom solid black curve (25) corresponds to $s = 250$ nm and the AQFP placement pitch in the x -direction equal to $S_x = S_q(w_q) + w_q$ determined by the 5% cross-coupling level between inductors L_q and L_{out} in the adjacent AQFPs in the same horizontal row. The dash black curve (23) corresponds to the n_{QFP} at S_x equal the physical width of the ac transformer $2l_L + s$ and $S_y = S_{min}(w, s)$. Red dash-dot curve (23a) corresponds to a hypothetical process with spacing $s = w$ in the excitation transformer and the AQFP cell size $2l_L + s$ determined by the transformer length; the number density saturates below $w_{min} = 50$ nm at $n_{QFP} = 20.2$ M per cm^2 , a level given by (23b). Red short-dash curve also corresponds to the case $s = w$, but with spacing between the inductors L_q set by a 5% cross-coupling level $S_q(w_q) = 3.92 \mu m$ according to the data in Fig. 3(b) (red dots).

bottom blue curves in Fig. 4. More importantly, these densities can be achieved at modest and already demonstrated linewidths and spacings, and only require implementing a kinetic inductor for L_q .

If a more aggressive process with $s = w$ is used, even higher values of n_{QFP} could be achieved as shown in Fig. 5 by the uppermost dash-dot curve, corresponding to (23a), because of a smaller cross-talk and smaller S_{min} between the ac transformers; see Fig. 3(b). At $w < w_{min}$, the n_{QFP} saturates at $(n_{QFP})_{max} = 2 \cdot 10^7 \text{ cm}^{-2}$, a level given by (23b) at $w = w_{min} = 50$ nm. This packing density could provide up to 6.7M MAJ3 logic gates per cm^2 and would be an $8 \times$ improvement over the standard SFQ5ee process. However, it may still be not sufficient for general purpose computing applications.

It appears that no significant increase in the circuit density is possible with microstrip inductors in planar transformers, mainly because of their strong cross-coupling. Accounting for other omitted components, e.g., JJs and interlayer vias, may only reduce the maximum densities estimated above.

D. AQFP Transformers Formed by Aligned Microstrips on the Vertically Spaced Planes

The area of AQFP ac excitation transformers can be decreased, at least by a factor of three, if microstrips forming the

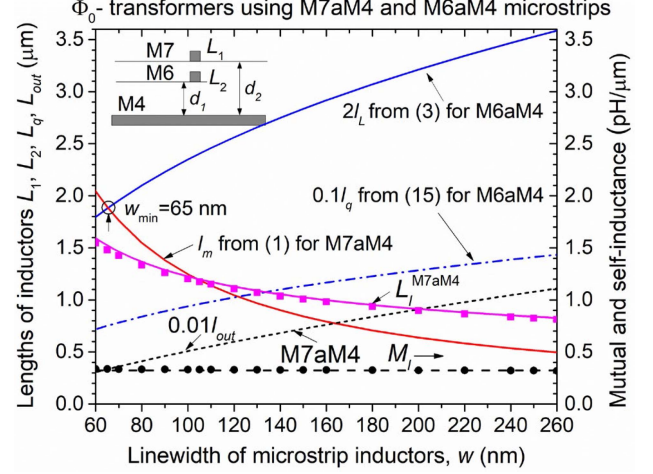


Fig. 6. Parameters of the ac excitation transformers using aligned equal-width microstrips M7aM4 and M6aM4 for the transformers primary and secondary, respectively. Inset: cross-section of the transformer. Top blue curve is the total length $2l_L$ (3) of two AQFP inductors L_2 (microstrips M6aM4) forming the transformer secondary, at $\beta_L = 0.2$, $I_{c,J} = 50 \mu A$. The bottom red solid curve is mutual running length l_m in (1) of the transformer primary (microstrip M7aM4) and the transformer secondary required to provide a Φ_0 flux excitation (dc + ac) in the secondary at the excitation current I_{max} (9) in the primary, assuming the maximum current density in Nb of $0.25 \text{ A}/\mu m^2$. Below $w_{min} \approx 65$ nm, $l_m > 2l_L$ and the required flux excitation cannot be provided without exceeding the critical current of Nb wire M7 in the transformer primary. Mutual inductance between the M7aM4 and M6aM4 microstrips per unit length M_l calculated using (26) at $d_1 = 615$ nm, $d_2 = 1015$ nm, $\lambda = 90$ nm, and $t_{M4} = t_{M6} = t_{M7} = 200$ nm is shown by the bottom black dash line; (•)—numerical simulations using wxLC and the same parameters. Self-inductance of M7aM4 microstrips per unit length, L_l^{M7aM4} , the sum of (13) and (7), is shown by the solid magenta curve; (•)—numerical simulations using wxLC [41]. Dash-dot blue curve is $0.1/l_q = 0.8/l_L$ (at $\beta_q = 1.6$) calculated using (15) and assuming that inductor L_q is a M6aM4 microstrip of the same width as the L_1 and L_2 . Black short-dash curve is the $1/100$ th of the length of inductor L_{out} calculated using (17b) and $M_q = M_l l_q$, assuming that it is a M7aM4 microstrip aligned over the L_q and having the of the same linewidth w .

transformer have the same width and locate above each other, over the same ground plane. In this case $A_{tr} = 2l_L w$.

If L_2 of the AQFP is the M6aM4 microstrips considered in Section III-A and III-B, the only convenient transformer primary is a microstrip M7aM4 with the signal trace on niobium layer M7. Mutual inductance per unit length of two microstrips with widths w_1 and w_2 , and thicknesses t_1 , t_2 , located on different planes is given in [32] by

$$M_l = \frac{\mu\mu_0}{4\pi} \ln \left[1 + \frac{4 \left(d_1 + \lambda + \frac{t_1}{2} \right) \cdot \left(d_2 + \lambda + \frac{t_2}{2} \right)}{p_x^2 + (d_2 - d_1)^2} \right] \quad (26)$$

where d_2 and d_1 are the dielectric thicknesses between the respective signal traces and the ground plane, and $p_x = s + \frac{w_1 + w_2}{2}$ is the horizontal distance between the geometrical centers of the microstrips' cross-sections. In the SFQ5ee process, $t_1 = t_2 = 200$ nm for both layers M6 and M7, $d_1 = 615$ nm, and $d_2 = 1015$ nm, corresponding to the interlayer dielectric thickness of 200 nm.

Mutual inductance (26) of the aligned microstrips M7aM4 and M6aM4, $p_x = 0$, is shown in Fig. 6 by the lowest black dash line along with the numerically simulated dependence shown by solid dots (•). Self-inductance per unit length of M7aM4 microstrips is given by the sum of (7) and (13).

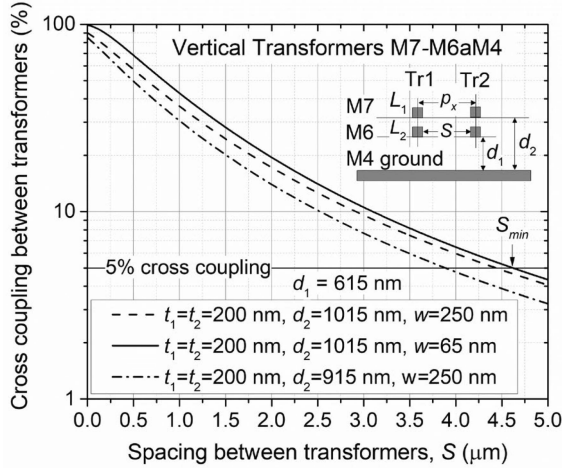


Fig. 7. Cross-talk between vertical transformers formed by the aligned equal-width M7aM4 and M6aM4 microstrips as a function of spacing between the transformers. Cross-talk is defined as mutual inductance between the primary of transformer #1 (Tr1) and the secondary of transformer #2 (Tr2) normalized to the mutual inductance inside the transformer. All curves were calculated using (26): solid curve corresponds to the parameters of the SFQ5ee process and $w = 250$ nm; dash curve is for $w = w_{\min} = 65$ nm; dash-dot curve corresponds to a hypothetical process with reduced to 100 nm dielectric thickness between layers M6 and M7, instead of 200 nm in the SFQ5ee. $S_{\min}$ is defined as the spacing at which the tolerable level of cross-coupling, set as 5% in this case, is reached. Dependence of $S_{\min}$ on w is very weak and an average value $S_{\min} = 4.53$ μm will be used to characterize the entire range of linewidths of interest $w_{\min} \leq w \leq 300$ nm.

Similarly to the planar transformers in Fig. 2, the required mutual running length of wires l_m in the “vertical” transformer needs to become longer than the two AQFP inductors L_2 (the transformer secondary) at $w_{\min} \lesssim 65$ nm because the sum of the ac and dc currents in the primary required to induce flux Φ_0 in the AQFP reaches the maximum allowed value $I_{\max} = \alpha j_c w_{\min} t = 3.25$ mA. The required transformer cannot be formed using narrower wires. The minimum cross-section $w_{\min} t_{M7} = 0.013$ μm^2 agrees perfectly with $A_{\min}$ estimated in Section II-A from the solution of (14).

Cross-talk between two parallel vertical M7-M6aM4 transformers is shown in Fig. 7 as a function of spacing S between them, for two linewidths $w = 250$ and 65 nm. Cross-talk is defined as the ratio of the mutual inductance between the M7 wire in transformer #1 and the M6 wire in transformer #2 to the mutual inductance in the transformer. The cross-talk very weakly depends on the w ; it reduces below the 5% level at $S \geq S_{\min} = 4.62$ and 4.44 μm , respectively, at $w = 65$ and 250 nm. Hence, instead of using a function $S_{\min}(w)$ as in Section III-B, we can simply use a single averaged value $S_{\min} = 4.53$ μm in the entire range of linewidths of interest. Overall, the cross-talk between the vertical transformers is larger than between the planar M6-M6aM4 transformers in Fig. 3 because d_2 is significantly larger than d_1 .

If L_q is an M6aM4 microstrip with the same width as the L_2 , its length l_q is the same as was described in Section III-B, Fig. 2, and $l_q > S_{\min}$. Hence, the AQFP number density only slightly differs from (22) and, neglecting cross-coupling in the same row

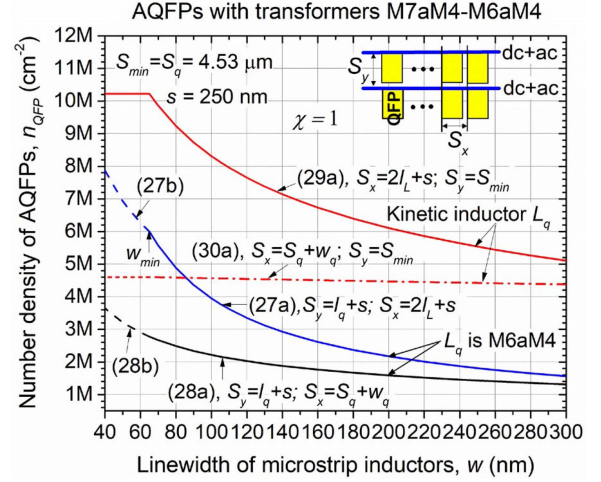


Fig. 8. Theoretical number density of AQFPs using vertical transformers formed by aligned, equal-width microstrips M7aM4 (transformer primary) and M6aM4 (transformer secondary) as a function of their linewidth. Inset: Top view of the AQFP cell placement along the ac power and dc flux bias transmission lines marked as dc+ac. The black solid and dash curves at the bottom correspond to Nb microstrips in the SFQ5ee process and the AQFP cell vertical spacing $S_y = l_q + s$ set by the physical height of the cell and the cell effective width $S_x = S_q + w$ set by the 5% cross-talk spacing S_q equal approximately 4.53 μm in the entire range of the linewidths. Blue solid and dash curves also correspond to Nb microstrips, but the cell placement pitch set by the physical width $S_x = 2l_L + w$, ignoring the cross-talk. The uppermost red solid and red dash-dot curve in the middle correspond to a hypothetical process, in which AQFP inductor L_q trace on the M6 level is made of a kinetic inductance material with width $w_q = 200$ nm to reduce its length below the critical cross-coupling distance $S_{\min}$ while preserving Nb microstrips as inductors L_2 in the ac excitation transformer.

of AQFPs, is

$$n_{\text{QFP}} = \frac{\chi}{(2l_L + s)(w + l_q + s)}, \text{ if } l_q + s \geq S_{\min}, 2l_L + s \geq S_q. \quad (27a)$$

This dependence is shown in Fig. 8 by the solid blue curve in the middle. At $w = 65$ nm and $s = 250$ nm, (27a) gives $n_{\text{QFP}} \approx 6 \cdot 10^6$ cm^{-2} , corresponding to about two million MAJ3 gates per cm^2 .

Using $w < w_{\min}$ for the inductor L_q reduces its length l_q . The AQFP number density in this case is similar to (22b) and given by

$$n_{\text{QFP}} = \frac{\chi}{(2l_L|_{w=w_{\min}} + s)(w_{\min} + l_q + s)}, \text{ if } w < w_{\min}, l_q + s \geq S_{\min}, 2l_L|_{w=w_{\min}} + s \geq S_q. \quad (27b)$$

This dependence is shown in Fig. 8 by the blue dash curve.

If cross-coupling between the AQFP outputs needs to be kept below a certain level, the adjacent inductors L_q in the same row need to be spaced at a safe distance S_q , and the AQFP number density reduces to

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w)(w + l_q + s)}, \text{ if } w \geq w_{\min}, l_q + s \geq S_{\min}, S_q + w \geq 2l_L + s \quad (28a)$$

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w)(w_{\min} + l_q + s)}, \text{ if } w < w_{\min}, l_q + s \geq S_{\min}, S_q + w \geq 2l_L + s \quad (28b)$$

$$\geq S_{\min}, S_q + w > 2l_L + s. \quad (28b)$$

These dependences are shown by the lowest black solid and dash curves in Fig. 8 for a 5% cross-coupling requirement $S_q = S_{\min} = 4.53 \mu\text{m}$, following from Fig. 7.

Since for all considered linewidths, $l_q + s > S_{\min}$, the y -direction size of the AQFP cell can be reduced by using a kinetic inductor for L_q to make $l_q + s \leq S_{\min}$. In this hypothetical process, the number density of AQFPs is similar to (23) and given by

$$n_{\text{QFP}} = \frac{\chi}{(2l_L + s)(w + S_{\min})}, \text{ if } S_{\min} \geq l_q + s, \quad (29a)$$

$$w \geq w_{\min}, 2l_L + s \geq S_q + w$$

$$n_{\text{QFP}} = \frac{\chi}{(2l_L|_{w=w_{\min}} + s)(w_{\min} + S_{\min})}, \text{ if } \quad (29b)$$

$$w < w_{\min}, S_{\min} > l_q + s, 2l_L|_{w=w_{\min}} + s \geq S_q + w.$$

These dependences are shown in Fig. 8 by the red solid curve saturating at $(n_{\text{QFP}})_{\max} = 1.02 \cdot 10^7 \text{ cm}^{-2}$, according to (29b).

And, finally, when the cross-talk is the factor in choosing the x - and y - placement pitches, the number density is given by

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w_q)(w + S_{\min})}, \text{ if } S_{\min} \geq l_q \quad (30a)$$

$$+ s, S_q + w_q > 2l_L + s, w \geq w_{\min}$$

and reaches the constant density at $w < w_{\min}$

$$n_{\text{QFP}} = \frac{\chi}{(S_q + w_q)(w_{\min} + S_{\min})}, \text{ if } \quad (30b)$$

$$S_{\min} \geq l_q + s, S_q + w_q > 2l_L + s,$$

$$w < w_{\min}.$$

These dependences are shown in Fig. 8 by the nearly horizontal red dash-dot and short-dash lines. The maximum number density in (30b) is $(n_{\text{QFP}})_{\max} = 4.4 \cdot 10^6 \text{ cm}^{-2}$ at the kinetic inductor L_q width $w_q = 0.2 \mu\text{m}$. This is a bit lower number than for the planar M6-M6aM4 transformers because of a larger $S_{\min}$ —the spacing at which the cross-coupling reduces to below 5%.

The n_{QFP} can be increased in a hypothetical fabrication process by using smaller thicknesses t_1 , t_2 , and d_2 to increase the mutual inductance in the transformer and decrease cross-coupling. For instance, at $t_1 = t_2 = 200 \text{ nm}$ and $d_2 = 915 \text{ nm}$, corresponding to the interlayer dielectric thickness between the M7 and M6 wires of 100 nm and $\Delta d = d_2 - d_1 = 300 \text{ nm}$ in (26), the $S_{\min}$ would decrease to about $3.8 \mu\text{m}$, as shown in Fig. 7.

To conclude, the achievable number densities of AQFPs using vertical microstrip-based transformers are comparable to those obtainable with the planar microstrip transformers in Section III-B because both types of the transformers have close values of the mutual inductance and similarly strong cross-coupling—the main drawback of using microstrips.

IV. STRIPLINE TRANSFORMERS AND AQFP NUMBER DENSITY

A. Stripline Transformers in the SFQ5ee Process

It is well known that cross-talk can be substantially reduced using stripline inductors with two ground planes instead of microstrips with one ground plane. The mutual inductance per unit length between two superconducting striplines is given in [32] by

$$M_l = \frac{\mu\mu_0}{4\pi} \ln \frac{\cosh \frac{\pi p_x}{H+2\lambda} - \cos \frac{\pi(h_1+h_2+2\lambda)}{H+2\lambda}}{\cosh \frac{\pi p_x}{H+2\lambda} - \cos \frac{\pi(h_2-h_1)}{H+2\lambda}} \quad (31)$$

where H is the dielectric thickness between the ground planes and $h_i = d_i + \frac{t_i}{2}$ is the distance between the bottom ground plane and the geometrical center of the cross-section of the i th signal wire, p_x is the in-plane (horizontal) distance between the geometrical centers of the stripline cross-sections. At large distances, M_l exponentially decreases with increasing p_x , with a decay length $p_0 = (H + 2\lambda)/\pi$.

In the existing SFQ5ee process, two parallel M6aM4bM7 (standing for M6 above M4 below M7) striplines can be used for a planar stripline-based transformer near the JJs. In this case, $H = 1015 \text{ nm}$ and $p_0 = 380 \text{ nm}$. Other possibilities would involve layers below the JJs and require multiple vias, leading to larger cell areas.

Because of a lower self-inductance and three times lower mutual inductance of the M6aM4bM7 striplines compared to the M6aM4 microstrips, lengths of the inductors L_2 and L_q , and the minimum width, $w_{\min} = 105 \text{ nm}$, shown in Fig. 9, are noticeably larger in this case than in Fig. 2. This leads to larger AQFP cell sizes and smaller n_{QFP} shown by a solid black curve in Fig. 10. However, *cross-coupling between the stripline-based transformers is dramatically lower than between the microstrip-based transformers*. Assuming a purely exponential decay, a 5% cross-coupling is expected at $S_{\min} \approx p_0 |\ln 0.05| = 1.14 \mu\text{m}$. The cross-coupling calculated using the full expression (31) at the in-transformer spacing $s = 250 \text{ nm}$ and numerical simulations give $S_{\min} = 1.34 \mu\text{m}$; see Fig. 11.

The number density of AQFPs using M6aM4bM7 striplines in Fig. 10 is given by (22a) and (22b) because $l_q \gg S_{\min}$ and $2l_L + s > S_{\min}$ in the entire range of w . Using $w < w_{\min}$ is possible for the inductor L_q , while keeping the excitation transformer linewidth at $w_{\min}$; this reduces the length l_q and the AQFP cell area. At $w = 60 \text{ nm}$, $n_{\text{QFP}} \approx 3.7 \cdot 10^6 \text{ cm}^{-2}$ and limited only by the cell size and not by the cross-talk. This is slightly lower a number than one can get using the M6aM4 microstrip inductors, for which the AQFP number density is limited by the cross-talk; see the lowest curve labeled 24b in Fig. 4.

If using a kinetic inductor, we could reduce the length of the inductor L_q to $l_q = S_{\min} = 1.34 \mu\text{m}$, the maximum number density would increase dramatically to $n_{\text{QFP}} \approx 1.8 \cdot 10^7 \text{ cm}^{-2}$, as shown in Fig. 10, whereas the in-row and between-rows cross-talk would remain below 5%. In the next section, we will consider the fabrication process required to achieve this.

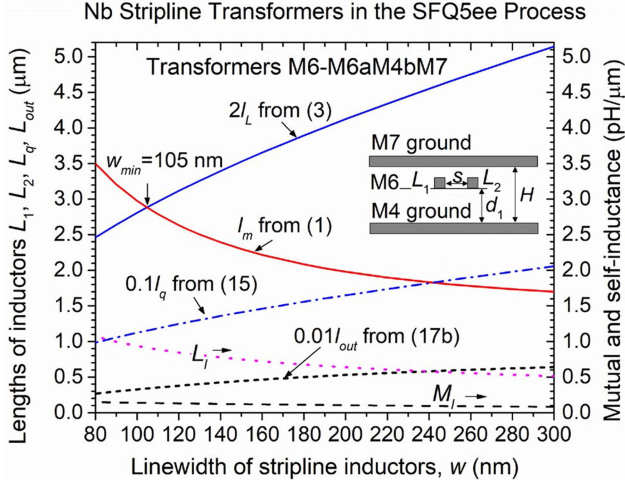


Fig. 9. Parameters of Φ_0 -excitation transformers using M6aM4bM7 striplines spaced at $s = 250$ nm in the SFQ5ee process with parameters: $d_1 = 615$ nm, $H = 1015$ nm, $\lambda = 90$ nm, and $t_{M4} = t_{M6} = t_{M7} = 200$ nm. Inset: cross-section of the M6-M6aM4bM7 transformers. Solid red curve is the mutual coupling length l_m in (1) of the striplines required to induce Φ_0 in the transformer secondary at the maximum possible current density in Nb of the transformer primary as a function of the striplines' linewidth. The uppermost solid blue curve is the length $2l_L$ of the transformer secondary, using (3) at $\beta_L = 0.2$, $I_{cJ} = 50$ μ A. The magenta dot curve is self-inductance of the M6aM4bM7 striplines per unit length, L_L , the scale is on the right axis; the bottom black dash curve is mutual inductance M_l per unit length between the striplines at $s = 250$ nm; and the scale is on the right axis. The blue dash-dot curve is $l_q/10$ from (15), which is equal $0.4l_L$, assuming that inductor L_q is a M6aM4bM7 stripline of the same linewidth as in the excitation transformer. The short dash black curve is $l_{out}/100$ calculated using (17b) and $M_q = M_l$, assuming that inductor L_{out} is also a M6aM4bM7 stripline spaced from the L_q at 250 nm. Below $w_{min} \approx 105$ nm, the primary current inducing the Φ_0 excitation at the coupling length $l_m = 2l_L$ exceeds 5.25 mA, the I_{max} of Nb wires. The assumed maximum excitation current density in the primary stripline is 0.25 A/ μ m².

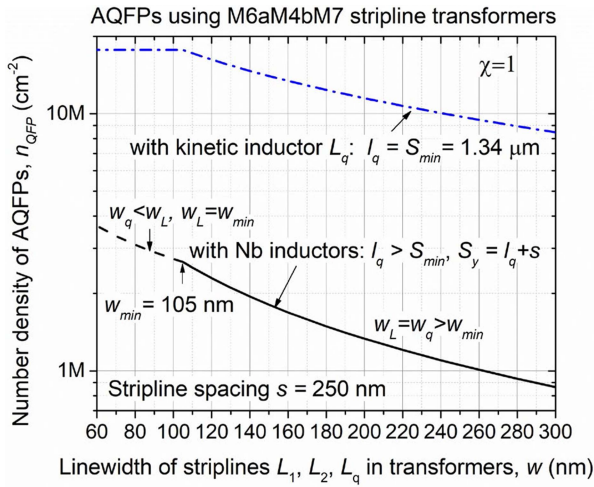


Fig. 10. Number density of AQFPs using striplines M6aM4bM7 spaced at $s = 250$ nm in all transformers as a function of striplines' linewidth. The bottom solid and dash black curves are for AQFPs using Nb striplines on the layer M6 in the SFQ5ee process with M4 and M7 ground planes at $H = 1015$ nm; below $w_{min} = 105$ nm, the linewidth of only inductor L_q , w_q , can be reduced to increase the AQFP density, as shown by the black dash curve. The top dash-dot blue curve is for a hypothetical process using Nb striplines in the ac excitation transformer L_1 , L_2 , and a kinetic inductor L_q , placed on the level of M6, with length equal to the 5% cross-talk length $S_{min} = 1.34$ μ m at $w = w_{min} = 105$ nm.

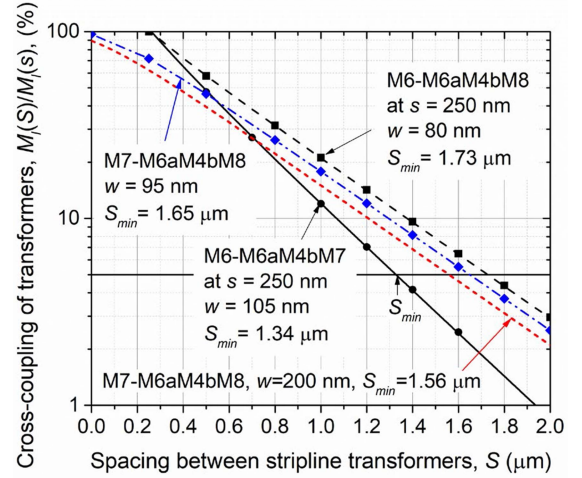


Fig. 11. Cross-coupling between stripline transformers as a function of spacing S between them: (•)—transformers formed by two striplines M6aM4bM7 spaced at $s = 250$ nm in the SFQ5ee process using M4 and M7 ground planes at $H = 1015$ nm, $w = w_{min} = 105$ nm; (■)—transformers formed by two striplines M6aM4bM8 spaced at $s = 250$ nm in the SFQ7ee process using M4 and M8 ground planes at $H = 1415$ nm, $w = w_{min} = 80$ nm, (♦)—transformers formed by a stripline M7aM4bM8 (dielectric thickness between M4 and M7 $d_2 = 1015$ nm, the SFQ7ee process) aligned over a stripline M6aM4bM8; width of both striplines $w = w_{min} = 95$ nm; red short dash is for the same striplines but with $w = 200$ nm. All the shown curves were calculated using (31), points are numerical simulation using wxLC [41]. In all the cases, we used $\lambda = 90$ nm, and thicknesses of the Nb layers $t_{M4} = t_{M6} = t_{M7} = t_{M8} = 200$ nm.

V. ADVANCED FABRICATION PROCESS SFQ7EE FOR SUPERCONDUCTOR LOGIC/MEMORY USING AC EXCITATION

A. Nb Stripline Transformers in the SFQ7ee Optimized Process

In order to utilize the advantages of a very low cross-talk between stripline transformers, small S_{min} , and substantially increase the number density of logic cells using flux transformers and ac excitation, we need to engineer an optimized fabrication process. From (31), the mutual inductance is maximized by decreasing the vertical distance between the signal wires $h_2 - h_1$ and placing them such that $h_2 + h_1 = H$, i.e., symmetrically in the middle between the ground planes at around $H/2$. These conditions could be nearly satisfied if we add a niobium layer M8 above the layer M7 in the SFQ5ee process. This would allow us to use M6aM4bM8 striplines in planar transformers and M7aM4bM8 striplines in vertical transformers. If we use 200 nm interlayer dielectric I7 (between M7 and M8), the distance between the ground planes M4 and M8 would become $H = 1415$ nm, giving the decay length $p_0 = 508$ nm. The corresponding fabrication process, titled SFQ7ee, is currently under development at MIT LL. Its cross-section is shown in Fig. 12.

Self- and mutual-inductances of Nb striplines M6aM4bM8 and M7aM4bM8 in the SFQ7ee process are shown in Fig. 13(a). Results of the analysis of transformers in the SFQ7ee process is shown in Fig. 13(b). For the planar M6-M6aM4bM8 excitation transformers in AQFPs, we get $w_{min} \approx 80$ nm at spacing between

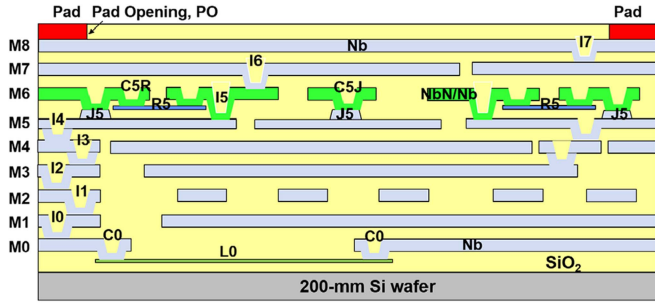


Fig. 12. Cross-section of an advanced process node SFQ7ee with 10 superconducting layers: layers M0–M4 and M7–M8 are Nb layers with 200 nm thickness; M5 is Nb layer, junction base electrode, with 150 nm thickness; M6 is NbN/Nb bilayer with 200 nm thickness; R5 is Mo or MoN_x resistor layer, 40 nm thickness; L0 is Mo₂N kinetic inductor, 40 nm thickness; Pad is Nb/Pt/Au contact pad metallization; I0–I4, I6, and I7 are vias in SiO₂ interlayer dielectric with 200 nm thickness; I5 via is in 280 nm thick dielectric. In comparison to the standard SFQ5ee node, SFQ7ee has an additional Nb layer M8. The layer M6 can be deposited either as a 200-nm Nb layer or as a NbN/Nb bilayer, e.g., with equal 100/100 nm thicknesses. In the latter case, 100-nm thick NbN kinetic inductors can be formed by selectively etching the top Nb layer of the bilayer, while the full bilayer can be used to form regular, low-value, geometrical inductors, as well as ac power and data transmission lines.

the M6 strips $s = 250$ nm and $w_{\min} \approx 55$ nm in the hypothetical case of $s = w$. In the former case, at $w = 80$ nm, $l_L = 1.52$ μm , $l_q \approx 12$ μm , and a 5% cross-talk is reached at $S_{\min} = 1.73$ μm ; see Fig. 11. We note that $S_{\min}$ slightly decreases with increasing the linewidth. Hence, we used the largest $S_{\min}$ corresponding to the $w_{\min}$ for the given type of the transformers.

If we use a stripline M7aM4bM8 as the transformer primary aligned over an equal-width stripline M6aM4bM8 secondary, the results are very similar: $w_{\min} \approx 95$ nm; at $w = 95$ nm, $l_L = 1.58$ μm , $l_q \approx 12.6$ μm , and a 5% cross-talk is reached at $S_{\min} = 1.65$ μm . Hence, in the SFQ7ee process with Nb inductors, the number density of the AQFPs using stripline inductors is limited only by the lengths of the inductors, especially by the inductor L_q , and given by (22a). It is shown in Fig. 14 by the bottom curves. In the practical range of the linewidth, $w \gtrsim 80$ nm, the AQFP number density is below $4 \cdot 10^6$ cm^{-2} .

B. Thin-Film Kinetic Inductors

The only way to significantly increase n_{QFP} , close to 18M per cm^2 , and utilize the advantages of striplines for lowering the cross-talk, is to reduce l_q down to $l_q \sim S_{\min}$ by using a kinetic inductor, as shown in Fig. 14 by the two top curves. To achieve this, the process layer stack would need a layer of kinetic inductors near the JJ layer, in addition to geometrical inductors and ac power transmission lines formed by Nb layers.

The first option for adding a layer of kinetic inductors near the JJ layer would be to replace the layer of resistors R5 in the SFQ5ee process stack if shunt resistors for JJs are not needed or can be moved below the layer of JJs as in the SC1 and SC2 processes [43], [11]. With this modification, the contact to the junctions' top electrode, J5, and to this kinetic inductor layer, K5, would be made by the layer M6 through vias, respectively, C5J and C5K, where label C5K would replace C5R in the existing SFQ5ee process. In case the shunt resistors are needed, they can

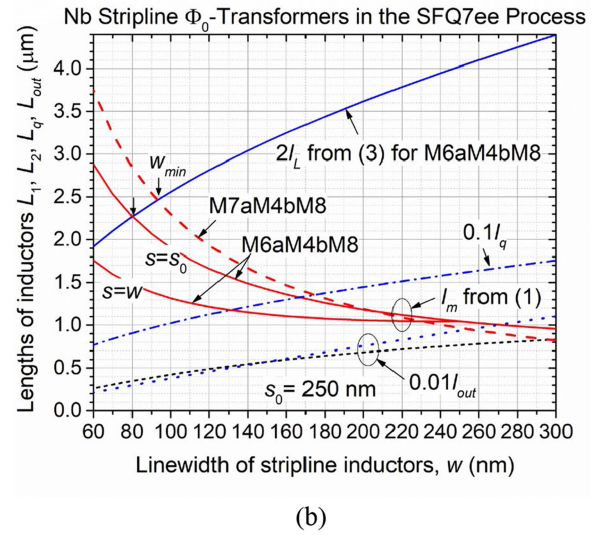
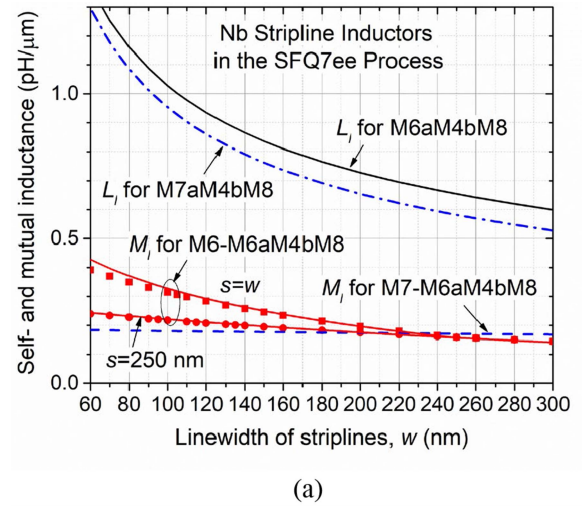


Fig. 13. (a) Self- and mutual-inductance per unit length of niobium stripline inductors in the SFQ7ee process nodes with 10 superconducting layers and parameters: $H = 1415$ nm (dielectric thickness between M4 and M8 ground planes); $d_1 = 615$ nm, dielectric thickness between M4 and M6; and $d_2 = 1015$ nm, dielectric thickness between M4 and M7. Thickness of all the Nb layers is 200 nm. Solid symbols (■) and (•) show the mutual inductance extracted using wxLC [41], the solid curves are the analytical expression (31). (b) Lengths of stripline inductors in the AQFPs: top solid blue line—length $2L_2$ (3) of the AQFP cell inductors L_2 if made as M6aM4bM8 stripline; dash red curve—length l_m (1) of the AQFP primary providing Φ_0 flux excitation if made as M7aM4bM8 stripline; solid red curves—length l_m (1) of the ac transformer primary providing Φ_0 flux excitation if made as M6aM4bM8 stripline spaced at either 250 nm from the transformer secondary L_2 (top red curve) or at $s = w$ (bottom red curve). Dash-dot blue curve is $0.1l_q$ (15), which equals $0.8 l_L$ if inductor L_q is the same M6aM4bM7 stripline as the L_2 . Short black dash and blue dot curves at the bottom are $0.01l_{\text{out}}$ (17b); l_{out} is the maximum length of the output inductor L_{out} if it is made as M7aM4bM8 stripline (blue dot) or as M6aM4bM8 stripline spaced from L_q at $s = 250$ nm (black short dash).

be moved into the position R4 below the layer M5, as in the SC1 process [43].

Since this K5 layer would replace a thin layer of resistors in the R5 position, only a thin film with similar thickness could be used, e.g., a 40-nm Mo₂N film used in the SFQ5ee process for bias inductors on the L0 layer. It has the sheet inductance of 8 pH/sq [36]. Another option would be a thin, 40 to 50 nm, NbN film

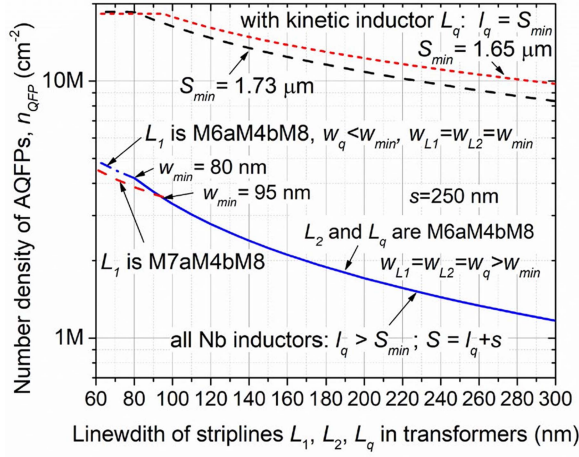


Fig. 14. Number density of AQFPs in the SFQ7ee process using Nb striplines M6aM4bM8 and M7aM4bM8 in the transformers (the bottom curves) and using a kinetic inductor for L_q (the top two curves), $\chi = 1$. Solid and dash-dot blue curves at the bottom correspond to all inductors made as M6aM4bM8 striplines spaced at $s = 250$ nm; below $w_{\min} = 80$ nm only the width of the inductor L_q can be reduced to decrease its length and increase the number density. The bottom red dash curve corresponds to the excitation transformer primary L_1 made as M7aM4bM8 stripline aligned over the equal-width secondary M6aM4bM8; below $w_{\min} = 95$ nm only the width of the inductor L_q can be reduced to decrease its length l_q and increase the number density. The top red short dash curve corresponds to the ac excitation transformers with M7aM4bM8 primary, having lower cross-talk ($S_{\min} = 1.65 \mu\text{m}$ at $w = w_{\min}$) than if the primary is a M6aM4bM8 stripline spaced at $s = 250$ nm from the secondary ($S_{\min} = 1.73 \mu\text{m}$ at $w = w_{\min}$). Both top curves assume that the length of the inductor L_q is adjusted to the value of $S_{\min}$ shown, the 5% cross-talk spacing, by using a kinetic inductor with the required sheet inductance.

with a similar sheet inductance [44]. Using these high kinetic inductance materials would result in very short inductors having very small mutual coupling to other inductors. For instance, at $w_q = 0.25 \mu\text{m}$, this would give $l_q \approx 0.33 \mu\text{m}$, which is too short a length to provide sufficient coupling to the output inductor L_{out} . Indeed, the strongest coupling to a K5 trace located at $d_1 = 505$ nm ($h_1 = 525$ nm) is achieved if the L_{out} trace is on the neighboring layer M6 at $d_1 = 615$ nm ($h_1 = 715$ nm). Then, using (31) or numerical simulations, we get the linear mutual inductance $M_{ql} \approx 0.32$ pH/ μm and the total mutual inductance $M_q = M_{ql}l_q \approx 0.1$ pH. According to (17b), this small mutual inductance would allow to transfer the output data only over distances below about $2 \mu\text{m}$.

Hence, for transferring the output data to larger distances, we need to use smaller sheet inductances in the range from 1.5 to 1.75 pH/sq determined by the largest desired length $l_q \approx S_{\min}$. Increasing thickness of the K5 layer significantly for achieving lower inductance is not desirable because this would require a full planarization of the K5 layer and increase the total dielectric thickness between the layers M5 and M6. The latter would make filling in of etched I5 vias by Nb of the M6 layer more difficult and reduce the via critical current. Therefore, we think that this approach is not practical and consider below another option.

C. NbN/Nb Bilayer of Kinetic/Geometric Inductors

The second option is to implement bilayer inductors—instead of the current 200-nm Nb layer M6 deposit a 200-nm NbN/Nb

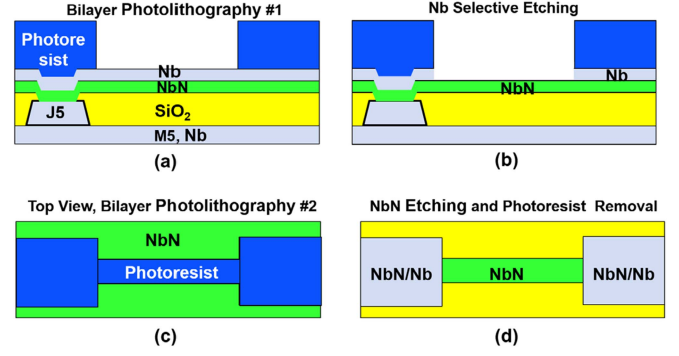


Fig. 15. Processing an *in-situ* deposited NbN/Nb bilayer, layer M6, to form kinetic inductors and geometrical inductors on the layer contacting Josephson junctions, J5: (a) cross-section of the structure after the first photolithography; (b) after selective etching of the top Nb layer, using a very thin etch-stop layer between NbN and Nb layers of the bilayer; (c) top view of the second photolithography defining kinetic inductors; and (d) final top view of the patterned composite inductor after NbN etching and photoresist removal. This inductor consists of a geometrical inductance part formed by the full bilayer and the kinetic part formed by the patterned NbN.

bilayer. By patterning individual layers of this bilayer independently, as shown in Fig. 15, we can create inductors in a very wide range of inductance values while maintaining an appropriate level of mutual inductance between them as explained below.

Inductance of a bilayer, L_{bi} can be calculated as a parallel connection of inductances of the individual layers L_{NbN} and L_{Nb} , assuming a sharp step-like change in the current density at the NbN/Nb interface

$$L_{\text{bi}} = \frac{L_{\text{NbN}}L_{\text{Nb}} - M_{\text{NbN,Nb}}^2}{L_{\text{NbN}} + L_{\text{Nb}} - 2M_{\text{NbN,Nb}}} \quad (32)$$

where $M_{\text{NbN,Nb}}$ is the aiding mutual inductance between the NbN and Nb layers of the bilayer. Our measurements [44] show that magnetic field penetration depth (London penetration depth) in reactively sputtered NbN films is approximately 490 nm, whereas it is 90 nm in the deposited Nb [11], [32]. Hence, in any practical range of the thicknesses of the individual layers, $L_{\text{NbN}} \gg L_{\text{Nb}}$ due to a much larger kinetic inductance of the NbN film, while geometrical inductances of both layers are nearly the same due to their close geometry and location. Also, $L_{\text{NbN}} \gg M_{\text{NbN,Nb}}$ because the mutual inductance is smaller than the geometrical inductance of each layer, and $M_{\text{NbN,Nb}} < L_{\text{Nb}}$. As a result, inductance of the bilayer (per unit length) is completely determined by the inductance of the top Nb layer, $L_{\text{bi}} \lesssim L_{\text{Nb}}$.

Dependence of inductance of the equal thicknesses, 100/100 nm, Nb/NbN bilayer M6 (biM6) per unit length calculated using (32) as a function of the bilayer linewidth is shown in Fig. 16 for biM6aM4aM7 and biM6aM4bM8 striplines along with the inductance of the top Nb layer of the bilayer. In the entire range of the linewidths, the bilayer stripline inductance is only about 2% lower than of the stripline using just the 100 nm top Nb layer of the bilayer. Hence, for all practical calculations $L_{\text{bi}} \approx L_{\text{Nb}}$.

A desired value L of an inductor with length l can be obtained by etching the top Nb layer from the bilayer over length l_{NbN} .

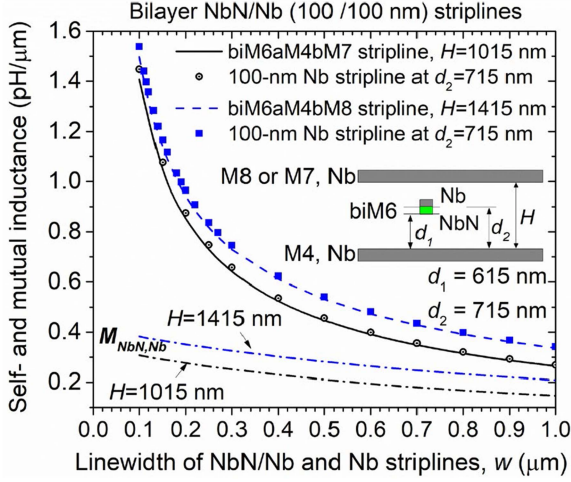


Fig. 16. Inductance per unit length of NbN/Nb bilayer, L_{bi} (thickness of the NbN and Nb layers is 100 nm) as layer M6 in striplines biM6aM4bM7 (solid black curve) and biM6aM4bM8 (blue dash curve) calculated using (32). Inset: cross-section of the bilayer striplines. Thicknesses of all layers correspond to the SFQ7ee process: $t_{M4} = t_{biM6} = t_{M7} = t_{M8} = 200$ nm and dielectric thicknesses are given in the inset. Inductance of the 100-nm Nb strip of the bilayer, L_{Nb} used in (32) is shown by open black and solid blue points. Mutual inductance per unit length between the layers of the bilayer, $M_{NbN,Nb}$ used in (32) is shown by the dash-dot curves for two dielectric thicknesses H between the ground planes, corresponding to the two types of the striplines. All self- and mutual-inductance values were extracted using wxLC software [41].

Neglecting inductance associated with electrical current redistribution between the bilayer and the bottom NbN layer near the ends of the etched Nb, the resultant inductance can be treated as a serial connection of the NbN (mostly kinetic) inductor and the full bilayer (mostly geometric) inductor, giving

$$L = l_{NbN}L_{NbN} + (l - l_{NbN})L_{bi}. \quad (33)$$

As was shown in [32], mutual inductance between two inductors with small cross-sections does not depend on their superconducting properties. Therefore, in the first approximation, mutual inductance between the partially etched bilayer and Nb stripline inductor on layer M7 can be represented as

$$M = l_{NbN}M_{NbN,M7} + (l - l_{NbN})M_{biM6,M7} \quad (34a)$$

and between the partially etched bilayer and the bilayer strip M6 as

$$M = l_{NbN}M_{NbN,biM6} + (l - l_{NbN})M_{biM6,biM6} \quad (34b)$$

where $M_{NbN,biM6}$ and $M_{NbN,M7}$ are mutual inductances per unit length between, respectively, a stripline in the bottom NbN layer of the bilayer and the parallel stripline in the full bilayer, and between the NbN stripline and Nb stripline inductor M7; $M_{biM6,biM6}$ and $M_{biM6,M7}$ are mutual inductances per unit length between, respectively, two striplines made of the bilayer, and between the bilayer stripline and the M7 stripline. All these mutual inductances can be easily calculated using (31) because they do not depend on superconducting properties of the signal strips [32].

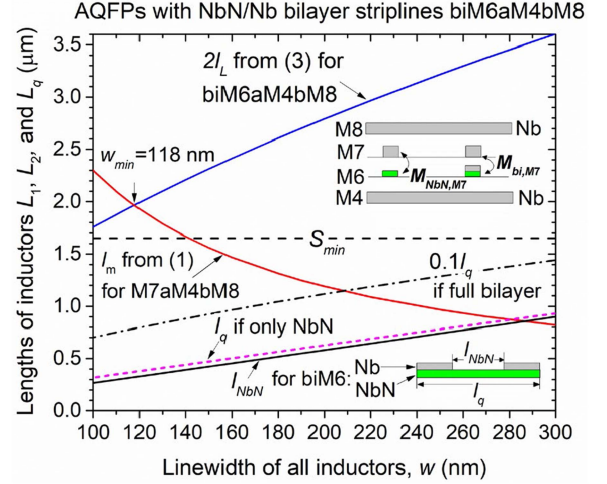


Fig. 17. Parameters of the ac excitation transformer and the AQFP cell inductors using Nb striplines M7aM4bM8 as the transformer primary and NbN/Nb bilayer striplines biM6aM4bM8 as the transformer secondary and the inductor L_q . Top Inset: cross-section of the stripline transformers using bilayers NbN/Nb (on the right side) and bilayers with etched Nb (on the left side). The assumed thicknesses of the NbN and Nb layers in the bilayer is 100 nm. Below $w_{min} = 118$ nm, the excitation current in the transformer primary stripline needs to exceed the critical current of the Nb wire of 6 mA in order to induce a total flux of Φ_0 in the secondary. The short-dash magenta line shows the length l_q of the inductor $L_q = 10.53$ pH if the inductor would be made completely out of the 100-nm-thick NbN film of the bilayer. The dash-dot black line shows $0.1l_q$ if the full length of the inductor is made of the bilayer. The bottom solid black line shows the length of the etched off Nb of the bilayer, i.e., the length l_{NbN} of the kinetic part of the bilayer inductor, see the bottom inset, required to make $l_q = 1.4$ μm so that $l_q + s = S_{min} = 1.65$ μm, the 5% cross-talk length.

D. AQFPs With NbN/Nb Bilayer of Kinetic/Geometric Inductors

Let us estimate dimensions of an AQFP using Nb stripline M7aM4bM8 for the ac excitation transformer primary, and NbN/Nb bilayer stripline biM6aM4bM8 for inductors L_2 (the secondary) and the output inductor L_{out} , and a patterned bilayer for the inductor $L_q = 10.53$ pH. Mutual inductance per unit length of the bilayer stripline and M7aM4bM8 stripline, $M_{biM6,M7}$ is the same as of Nb striplines M6aM4bM8 and M7aM4bM8 shown in Fig. 13(a) because they have the same locations and geometrical dimensions; see (31) and [32]. Therefore, dependence of the required mutual length on the linewidth, $l_m(w)$ in (1) is also the same as shown in Fig. 13(b) by the red dash curve. Using inductance per unit length of the 100/100 nm NbN/Nb bilayer, we calculate the total length of the excitation transformer secondary, $2l_L(w)$ shown in Fig. 17. The minimum linewidth of the primary determined from the condition $l_m(w) = 2l_L(w)$ is $w_{min} = 118$ nm, the largest value in all the considered cases due to the smallest value of $2l_L$ which in turn is a result of a higher inductance of the biM6 layer than of the 200-nm Nb M6 layer in all the other cases. In the entire range of the linewidths, $2l_L + s \gg S_q = S_{min}$ for the stripline transformers; see Fig. 11. So the cross-talk in the same row of AQFPs can be ignored.

The l_q can be adjusted to any desired lengths between the shortest, $l_q/L_{NbN}(w)$, and the longest lengths, $l_q/L_{biM6}(w)$,

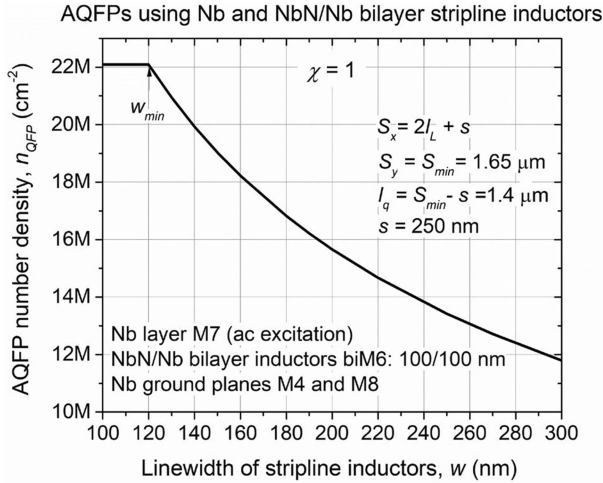


Fig. 18. Theoretical number density of AQFPs using NbN/Nb bilayer and Nb stripline inductors in the proposed advanced version of the SFQ7ee process; see text. For these AQFPs, the placement pitch in the x -direction, S_x is set by the physical width of the AQFP cell $2l_L$ and the minimum spacing s used in the process, because it is larger than the cross-talk length $S_q = S_{\min} = 1.65 \mu\text{m}$. The y -direction placement pitch S_y of the AQFP rows is set equal the 5% cross-talk spacing $S_{\min} = 1.65 \mu\text{m}$. This sets the length l_q of the AQFP inductor L_q which is made by etching Nb off the NbN/Nb bilayer on length l_{NbN} shown in Fig. 17.

as shown in Fig. 17, respectively, by short-dash magenta and dash-dot black lines; here $L_{\text{NbN}}(w)$ and $L_{\text{biM6}}(w)$ are width-dependent linear inductances of, respectively, the NbN layer and of the NbN/Nb bilayer. This l_q adjustment can be done by etching the top Nb layer of the bilayer on its full length or only on a part l_{NbN} of it; see the bottom Inset in Fig. 17. To minimize cross-talk between the adjacent rows of AQFPs, we need $l_q + s \leq S_{\min}$. For the vertical stripline transformers, $S_{\min}$ weakly depends on w and $S_{\min} = 1.65 \mu\text{m}$ at a practical minimum width of the bilayer inductors $w = 100 \text{ nm}$; see Fig. 11, blue dash-dot and red short dash curves.

The required Nb etch length to get the composite inductor with $L_q = 10.53 \text{ pH}$ and $l_q = 1.40 \mu\text{m}$ (at $s = 250 \text{ nm}$), i.e., the length of the kinetic part l_{NbN} is shown in Fig. 17. This etch length is certainly within the capabilities of the existing SCE fabrication technology. Making shorter l_q by etching a larger length of Nb off the bilayer strip is certainly possible. This would decrease the cell size but increase the cross-talk and also decrease the data output length l_{out} . In practice, the actual length will be a design-dependent and adjustable parameter.

Fig. 18 shows the theoretical number density (29a) and (29b) of the AQFPs using the proposed fabrication process with the bilayer inductors. As can be seen, densities above 22 M cm^{-2} , corresponding to over 7 M MAJ3 gates per cm^2 , can be reached at very modest linewidths of all the inductors, which are fully within capabilities of the existing fabrication technology. This is the main result of this section.

VI. DISCUSSION

We have considered two main factors limiting the integration scale—device number density—of superconductor integrated

circuits using ac power for logic and/or memory cell excitation and clocking: critical current of superconducting transformers and their cross-coupling. We used parameters of the fabrication processes developed at MIT LL and parameters of the AQFP cells as an example for detailed numerical simulations. Below we discuss how our general conclusions depend on the selected fabrication process, the typical critical current of the cell JJs I_{cJ} , the type of superconductor logic, and so on.

From (14), the minimum cross-sectional area and the minimum linewidth of ac power delivery transmission lines are directly proportional to, respectively, I_{cJ}/β_L and $(I_{cJ}/\beta_L)^{1/2}$ regardless of the cell type. In this respect, cells using smaller critical currents and larger β_L than the considered AQFP cells, e.g., RQL cells, allow for smaller $w_{\min}$ than in the AQFP cells. However, the cell area is proportional to the β_L . So logic types using larger β_L values, larger than 1.6 in AQFPs, e.g., RQL [25], [26], [27], would in general have lower densities than the densities estimated in Sections III–V for the AQFP cells.

The wave impedance of superconducting striplines $Z_0 = (L_l/C_l)^{1/2}$ grows rapidly with decreasing the linewidth due to the growing kinetic inductance and diminishing capacitance, and becomes larger than 50Ω at $w \lesssim 250 \text{ nm}$, see Fig. 4 in [11], where C_l is capacitance per unit length. We assumed that impedance matching to the ac power source can be done off-chip, whereas the standard design practice is to use a fixed-width 50Ω transmission lines on the chips. The latter only decreases circuit density compared to the given estimates.

Increasing I_{cJ} increases the circuit density until the maximum density is reached at some I_{cJ}^{opt} at which the area occupied by the circuit JJs is approximately equal to the area occupied by the circuit inductors. At $I_{cJ} > I_{cJ}^{\text{opt}}$ the density decreases. The I_{cJ}^{opt} weakly increases with increasing the process Josephson critical current density. However, increasing I_{cJ} proportionally increases power dissipation in the circuit junctions and also increases ac power loss caused by dielectric losses in the transmission lines, which grows as I_{cJ}^2 because the ac excitation current grows proportionally to I_{cJ} . As a result, selection of I_{cJ} is made based on the total energy dissipation requirements for the circuit and on the tolerable level of the circuit BER, not on the circuit density optimization.

Since magnetic inductance and mutual inductance of the microstrip and stripline inductors depends logarithmically on the linewidth, signal trace thickness, and dielectric thickness, our main results weakly depend on the exact parameters of the fabrication process selected for the analysis. Moreover, the main parameters of different fabrication processes are very similar to the MIT LL processes considered; see, e.g., [45].

A. Scaling Problems With AC Power Distribution and AC Power Dissipation

Yet another problem impeding scaling up ac-powered superconductor digital circuits is delivery of multiphase ac power and associated on-chip power losses. To illustrate severity of this problem, let us estimate ac power requirements and dielectric losses in a large-scale integrated circuit containing, say, $N = 10^7$ AQFP cells. Let us use a conservative linewidth of the ac

transformer primary stripline M7aM4bM8 of 200 nm in the most advanced SFQ7ee process which, at this linewidth, could provide AQFP density of $1.6 \cdot 10^7 \text{ cm}^{-2}$ at below 5% cross-talk; see Fig. 18. The length of the AQFP cell along the ac power transmission line is $2l_L = 2.8 \text{ } \mu\text{m}$; see Fig. 17. The total length of the ac power transmission lines required to deliver ac power to all the AQFP cells is $l_{ac} = (2l_L + s)N = 30 \text{ m}$. This is 150 times larger than the ac wavelength, $\lambda_{clk} \approx 20 \text{ cm}$ at the typical clock frequency, f_{cl} , of 5 GHz.

Delivery of the, typically four-phase, ac excitation/clock current from one or multiple sources as a traveling wave is not possible because different AQFPs along the transmission lines will be experiencing different phases due to propagation delays. The only option is to create a power distribution tree, a power grid, with all branches much shorter than $\lambda_{clk}/10 \approx 2 \text{ cm}$ which, hopefully, can work as lumped elements maintaining the same ac phase along the branch and the required phase difference between the different branches. The number of the branches and the required ac power splitters is equal to the number of AQFP rows, or l_{ac}/a , where a is the row length; at $a = 1 \text{ cm}$, this number is 3000.

At $w = 200 \text{ nm}$, stripline M7aM4bM8 has $Z_0 = 62.5 \text{ } \Omega$ and $C_l = 168.1 \text{ pF/m}$, based on the relative dielectric constant $\varepsilon = 4.6$ for the SiO_2 interlayer dielectric in our fabrication processes [46] and capacitance extraction software [41]. The total capacitance of the power grid is $C = C_l l_{ac} = 5.04 \text{ nF}$. The ac current amplitude in the AQFP primary under consideration, I_{ac} , creating a $\Phi_0/2$ peak excitation in the AQFP secondary wire is about 2.13 mA. Hence, the ac peak voltage on each branch of the power tree is $V_{ac} = I_{ac} Z_0 \approx 0.133 \text{ V}$. Since each branch needs 2.13 mA peak current, the total peak current I_{tot} in the tree trunk is $3000 \times 2.13 \text{ mA} \approx 6.4 \text{ A}$. Hence, the total ac power required to be delivered to and taken away from the chip should be $P = \frac{1}{2} I_{tot} V_{ac} = 0.43 \text{ W}$.

The peak energy stored in the grid is $E = CV_{ac}^2/2 = 4.46 \cdot 10^{-11} \text{ J}$. The ac power loss due to dielectric losses in the grid capacitor C is $P_{loss} = 2\pi E f_{cl} \tan \delta \approx 2.1 \text{ mW}$, where the dielectric loss tangent for our low-temperature deposited SiO_2 is $\tan \delta \approx 1.5 \cdot 10^{-3}$ [46]. At the same time, dynamic power consumption due to adiabatic switching of the AQFP cells is $P_d = N E_{sw} f_{cl} = 50 \text{ nW}$, based on the AQFP switching energy of $1 \cdot 10^{-21} \text{ J}$ [22], [34], [47]. *So the parasitic dissipation in the power grid is 42 times larger than the useful power dissipation related to the AQFP operation. As a result, the real AQFP circuits have much lower, about 42 times lower, energy efficiency than what is typically claimed [47] based solely on the switching energy consideration.*

On-chip power dissipation of 2.1 mW at 4 K is equivalent to 2.1 W power consumption at room temperature because the cryocooling penalty is about 1000 W per W [15]. This total power consumption by a 10-million-devices, 3.3 million gates, chip is comparable to or even larger than the power consumption by CMOS chips with similar complexity.

VII. CONCLUSION

For each type of superconducting transformers, there is a minimum cross-sectional area $A_{min} \sim 1.5 \cdot 10^{-2} \text{ } \mu\text{m}^2$ and the

corresponding minimum linewidth $w_{min} \sim 0.1 \text{ } \mu\text{m}$ of the ac power transmission line, the transformer primary, below which the superconducting current providing flux excitation required for the cell operation exceeds the critical current of the wire. This critical current as well as the mutual inductance set the minimum mutual coupling length between the transformer primary and secondary and hence the minimum size of the logic or memory cells in the x -direction, along the ac power line. On the other hand, reduction of the linewidth of the transformer secondary L_2 increases its kinetic inductance and, hence, decreases its length since the total inductance is set by the cell design parameter β_L . This length reduction decreases the mutual coupling in the transformer and prevents transformer miniaturization.

Mutual coupling (cross-talk) between adjacent transformers is strong and long-ranged if transformers use microstrip inductors. This presents a serious problem and limits the scale of integration. Cross-talk diminishes exponentially with spacing between the transformers if stripline inductors are used instead of microstrips. However, mutual coupling of striplines is much weaker than of the microstrip, which increases the size of the stripline-based transformers.

Using parameters of AQFP cells as a typical example, we have estimated the maximum number density of AQFP circuits for all types of microstrip and stripline inductors which can be formed near the JJs in fully planarized fabrication processes for superconductor electronics developed at MIT Lincoln Laboratory. We have shown that, at the SFQ5ee process minimum linewidth and spacing $w = s = 250 \text{ nm}$, the theoretical AQFP number density is about 1 M per cm^2 . Reduction of Nb linewidth to about 60 nm in the future processes may increase the AQFP number density to a few million per cm^2 .

We have shown that the circuit density can be substantially increased by using kinetic inductors, e.g., patterned NbN films instead of Nb, mainly geometrical, inductors. Since short strips of kinetic inductors have very small mutual coupling, we have proposed to use bilayer inductors, e.g., NbN/Nb bilayers consisting of a layer of kinetic inductor (material with large λ) covered by a layer of geometrical inductor (material with small λ , niobium). Partial patterning of the top Nb layer of the bilayer enables making kinetic inductors with a wide range of inductance values from the patterned bottom layer, whereas using the full bilayer allows for making small-value inductors and preserves sufficient mutual coupling.

Additional design flexibility is provided by selecting thicknesses of the individual layers in the bilayer. As an example, we have considered a 100/100 nm NbN/Nb bilayer as the layer M6 in a future fabrication process node SFQ7ee having nine superconducting layers, three of which are above the layer of JJs. We have calculated parameters of the transformers and dimensions of the AQFP cells using Nb striplines M7aM4bM8 for ac power delivery and striplines biM6aM4bM8 with patterned Nb of the bilayer for the AQFP cell inductors. We have found that the proposed advanced fabrication process with bilayer inductors allows for the highest AQFP number density among all the considered processes and options, reaching above $22 \text{ M AQFPs per cm}^2$, corresponding to about $7 \text{ M MAJ3 logic gates per cm}^2$, at modest linewidths $w \gtrsim 120 \text{ nm}$, which are within the

capabilities of the fabrication equipment and existing fabrication processes for superconductor electronics.

The current status of the SFQ7ee process and NbN/Nb bilayer inductors will be published elsewhere [48].

Although all the calculations were done for the typical parameters of AQFP cells, the same conclusions and scaling estimates, with small modifications, are applicable to all other circuits using ac power and/or transformers such as RQL, nSQUID circuits [4], ac-biased SFQ circuits with ac/dc converters [49], and circuits with SFQ biasing [50], and likely to neuromorphic bioSFQ circuits [51] using flux transformers in artificial superconducting neurons.

The proposed bilayer inductors could benefit miniaturization of all superconductor digital circuits, especially those using small values of I_{cJ} and correspondingly large values of cell inductors.

Energy efficiency of adiabatic and nonadiabatic ac-powered superconductor digital circuits is limited by dielectric losses in the ac power transmission lines. Superconductor ac-powered circuits can be made overall significantly lower power consuming than CMOS circuits only if we either implement interlayer dielectrics with loss tangent $\tan \delta \sim 10^{-5}$, which requires development of a new fabrication technology, or significantly reduce the ac excitation/clock frequency, to about 0.1 GHz, or both.

The estimated maximum cell number density of ac-powered circuits of about $2 \cdot 10^7 \text{ cm}^{-2}$ should be considered as the upper limit to the achievable scale of integration of superconductor electronics utilizing inductively coupled ac excitation and clocking.

Capacitive coupling of RQL cells to the power grid, instead of inductive coupling, was patented a few years ago [52], despite that capacitors have been used in ac power distribution circuits during at least the last 150 years and that most of the superconducting qubit circuits have been using capacitive coupling to the ac transmission lines for the last 20 years. It has not been demonstrated for RQL and AQFP cells. Scaling limitations of this approach require a separate discussion.

A separate discussion is also required of potential applications of superconductor ac-clocked logics and memories for which the scale of integration and performance estimated in Sections III–VI would be sufficient. For instance, whether it is sufficient for applications in general purpose or high-performance computing, data centers, artificial neural networks and neuromorphic processors, cold processors for quantum computers, cold processors for large arrays of cold sensors, etc.

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